

WinISD Pro Help

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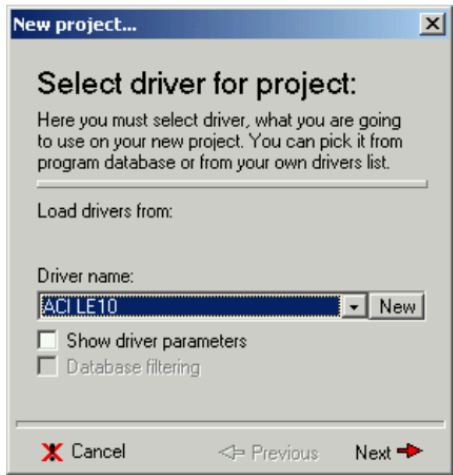
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Getting Started

Selecting a Driver and choosing a Box Type:

Start by loading a driver into WinISD as follows: Click on File, then New Project (or, simply click on the New Project icon, which is the second icon from the left on the Tool Bar). The New Project window (below) is presented.



Click on the down arrow in the drop down menu box, and you will be able to scroll through and choose a driver from the Program (WinISD) database (see Creating Your Own Driver Database for details on how to Create and Manage your own personal database). Putting a check mark in the 'Show driver parameters' box will cause WinISD to show you the [Thiele/Small](#) parameters of the driver that you have selected. This way you can see whether or not this particular driver is what you're looking for before you load it into WinISD as shown below. Note that you can change between standard and metric measurements simply by clicking on the desired measurement system (mm, in, etc..). You can also change any value the same way.

New project...

Select driver for project:

Here you must select driver, what you are going to use on your new project. You can pick it from program database or from your own drivers list.

Load drivers from:

Driver name:

ACI LE10

New

☒ Show driver parameters

☐ Database filtering

X

Cancel

Previous

Next

Qts: 0,490

Qms: 6,762

Vas: 37,85 l

Qes: 0,529

Fs: 35,67 Hz

SPL: 87,15 dB

Re: 1,488 ohm

Pe: 200,0 W

Le: 0,00 mH

BL: 7,16100 Tm

Xmax: 6,0 mm

Dd: 205,0 mm

Z: 2,000 ohm

Sd: 330,0 cm²

Choose a driver to work with (for demonstration purposes, choose the JL Audio 10W1-4 driver...you can either scroll down the menu until you find it, or you can type the beginning letters "JL..." and the window will "scroll to" accordingly. Click NEXT, and the Driver Usage window will appear:

New project...

Driver usage:

Here you need to enter number of drivers used to your new project and select placement type for drivers.

Drivers:

1

Placement:

☒ Normal

☐ Iso-barik (Compound)



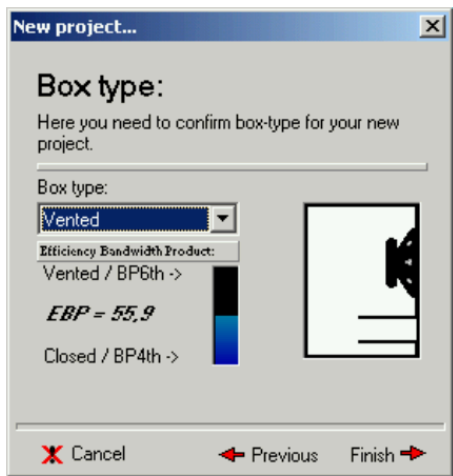
X

Cancel

Previous

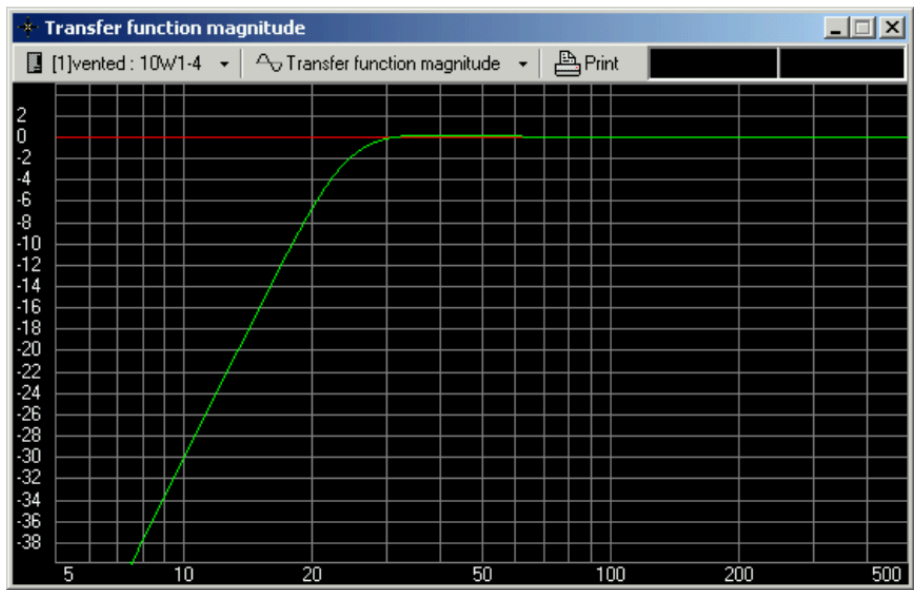
Next

Here is where you decide whether you're going to design the driver(s) to be configured as Standard or Isobarik. You can also choose the number of drivers you are going to put into the box. Don't worry, all of these parameters can be changed "on the fly", even after the driver is loaded (plotted on the graph). If you made a mistake and chose the wrong driver on the previous screen, click on PREV to go back to the New Project window. For demonstration purposes, choose 1 driver in a Normal configuration (placement). Click on NEXT to bring up the Box Type window:

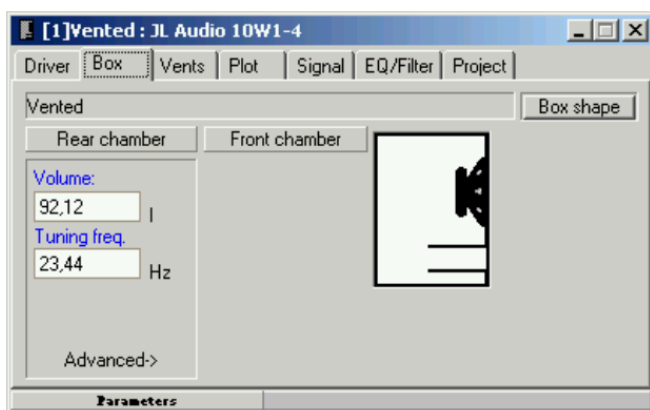


Here, you can choose whether you are planning to build a Sealed, Vented, or Bandpass box (these are referred to as the box types). Use the vertical Efficiency Bandwidth Product (EBP) bar indicator as a guide to determine which box type is best suited for this particular driver. If the bar is high, then this driver is best suited for installation in a vented box (or a 6th order bandpass). Conversely, if the bar is low, go with a closed type box (or a 4th order bandpass). If the bar is near the center (as it is here), then the driver will likely perform equally well in either a closed or vented type box.

The choice you make here is not changeable later on but you can, of course, load this driver simultaneously multiple times in different box types. Choose a box type by clicking on the down arrow next to the drop down menu box, and click on the desired box type. For demonstration purposes, choose Vented. Click FINISH, and you will then be presented with the Plot Window, with the plot of the JL Audio 10W1-4 driver as seen below. The "plot" is the curved (green,in this example) line:



Also displayed on your screen will be the Project Window (below). Position the Project Window to a comfortable position, and then click on the Box tab of the Project Window, which will now display the Volume and Tuning Frequency of the box:



Clicking on the 'Parameters' button will open a drop down menu which will show you the [Thiele/Small](#) specs of the driver, and will also allow you to make temporary changes to those values and their system of measurement (as outlined in the first paragraph above). The green color bar corresponds to the green plot on the plot window (clicking on the green bar will open a window which will enable you to change the color of the plot, if you so desire).

What the information here in the box tab is telling you is that in order for the JL Audio 10W1-4 woofer to produce the sound that is indicated by the green plot in the plot window, it must be placed in a 90.30 liters box, and that this box must be "tuned" to 23.81 Hz (vented or bass-reflex boxes are "tuned" by the number of, diameter of, and length of the "ports", which are usually tubes of PVC pipe built into the box and "vented" to the outside air, as demonstrated by the two horizontal lines in the diagram below the words "Box Shape" in the Project Window).

The initial green plot shown, as well as the initial Box Volume and Tuning Frequency data (in the Project Window) shown, gives us the smoothest calculated attainable plot for this driver. Notice there are no "peaks" or "dips" in the green plot. The green plot is the ACTUAL REPRESENTATION OF HOW THIS SPEAKER WILL SOUND IN A COMPLETED BOX.

WinISD will always default to show you the smoothest (well, at least quite close) attainable plot when you first load a driver (sometimes, if a driver that was designed to be installed into a sealed box is loaded into WinISD in a vented box configuration, the initial plot will look like a roller coaster...however this doesn't necessarily mean that this driver won't work in a vented box, since some drivers are designed to work in both types of boxes). Some drivers have very low Qts values. With them, it might seem that they can't reproduce any bass in default box. However, you can adjust the box as you watch the [transfer function magnitude](#)-plot. Notice that I did not say that the initial plot is the "best" or "ideal" plot, only that it is the smoothest plot. The reason I avoided those words will soon become clear.

You can change the plot (which, of course, will change the way the speaker sounds after it's built), by changing the Volume and/or the Tuning Frequency of the box. If you lower the Volume of the box (which makes for a smaller box), you must give back something in exchange, the price here being low frequency response.

Notice that the green plot crosses the -3dB horizontal line at 23 Hz (left clicking in the plot window and dragging the mouse will activate the Trace Cross hairs, give it a try) . This purple line is the -3dB (or half-power, or cutoff) point. This line represents the frequency at which the usable bass output of the box "runs out of steam". Any frequencies below the intersection at the -3dB point simply won't be reproduced very well by the box. In our case here, that frequency is 23 Hz. Most low bass speakers are designed so that the -3dB point is somewhere around 35 Hz. Since we are dealing with a particularly well designed (expensive) woofer, we are subsequently being blessed with some room to play with! We are in the enviable situation of having to RAISE the -3dB point of the box! Since this is done by LOWERING the Box Volume, we're in a win/win situation (remember, as I said, this is an expensive, high quality woofer we're dealing with here, so don't expect to run over to Joe's Flea Market and find such a woofer).

Now, let's say that we want this box to cutoff around 35 Hz. Double-click on the numbers below the words volume, and change the volume to 1.5. Also change the Tuning freq: to 30 the same way. This, of course, tells WinISD to recalculate the woofer when placed in a 1.5 cubic foot box tuned to 30Hz, and to display the corresponding plot. Note that this is less than HALF the initial given box size...we've traded some low bass response (that we didn't even need in this case...remember we went from a cutoff of 23 Hz to a cutoff of to 32 Hz) and ended up with a much smaller box!

By lowering the box volume, we've brought up the cutoff point of the box. That's a nice curve, albeit a challenging box to build due to the small enclosure and LOW tuning frequency (rectangular vents are probably in order).

"Ok, that's all nice and fine, but how did you know what numbers to plug into WinISD to achieve this plot?", you ask. Well, here's where you're gonna learn about the true power of WinISD, my friend. Position your mouse cursor over that diagram of the speaker (below the words Box Shape) in the Project Window (under the Box tab, if you've gone astray). Click and hold your left mouse button. Move your mouse. "Draw" the plot that you want. Release the rodent. Observe the new displayed Box Volume and Tuning Frequency data. Now, wasn't that easy? Want a little secret about drawing your plot? Click [here](#).

If you like the looks of the plot, then run out to the garage, build your box, and you're done. Congratulations, you've just

designed and built your first speaker which in this case, by the way, would be a very awesome speaker. And all this made possible by this kick-ass (sorry, kids) little freeware program called WinISD!

All of the features we've covered here so far and more are covered in detail in the [Designing your Box](#) Help section. Go ahead and load some more various drivers and play around with the plots (to change the color of a plot, click on the colored bar at the bottom of the Project Box). You'll soon see what I mean about the quality of the driver we were just working with. There are other manufacturers who specialize in manufacturing drivers as serious as JL Audio's (these drivers are designed mainly for car audio competition), but not many.

Many lower-cost drivers perform admirably in their own right, however, and it's not always necessary to spend the big bucks to get a great sounding speaker, especially if you're willing to settle for a larger box! Now that you have the ability to actually "build" the box before you buy the driver, you can actually take different drivers for a "test drive" before you open your wallet. Search around on the net, and just plug the Thiele/Small specs of any drivers that interest you into WinISD and check out the plot vs. box size. Sooner or later, you'll find that "perfect" driver for your application, and all without ever leaving the house or dealing with salesmen!

Note that the author of WinISD (nor myself or any others on the WinISD team) are not in any way affiliated with JL Audio, or any other speaker manufacturer. I simply chose to use the 10W1-4 for the above help demonstration to illustrate a point.

The old adage "Haste makes Waste" applies more to speaker building than perhaps anything else on earth! From the very first speaker building thought in your mind all the way to the end product, haste will most certainly make waste. And alot of waste, at that. Each and every step along the way must be carefully planned/thought out as you go. "Fruition Day" will most certainly underscore your faults more than your achievements. I hate to sound so pessimistic, but you need to approach the speaker design task in the correct frame of mind. Don't cut corners while mentally thinking "ah, it'll be alright, nobody will know". This thinking will at best result in a box that YOU know isn't performing at its best. And that thought will haunt you every time you crank it up. If you're gonna invest the time and energy to do it in the first place, then by all means, take the time to do it right!!!!

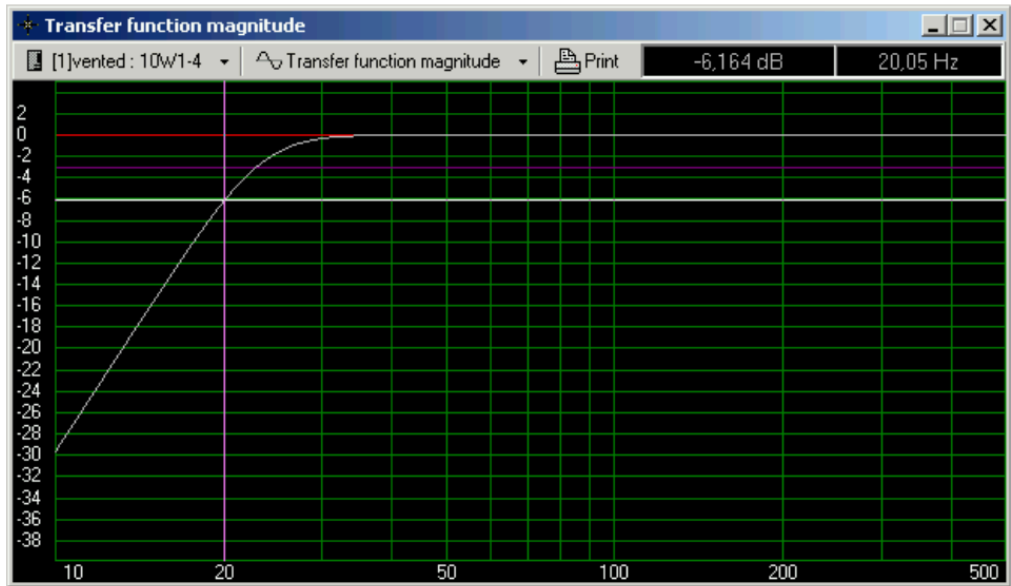
This page originally written by JJ Richard (JJRichard@linearteam.org) and revised by Janne Ahonen (janne@linearteam.org)

Designing your Box (a.k.a. Designing your Project)

This section of the Help File assumes that you've read the [Getting Started](#) section first.

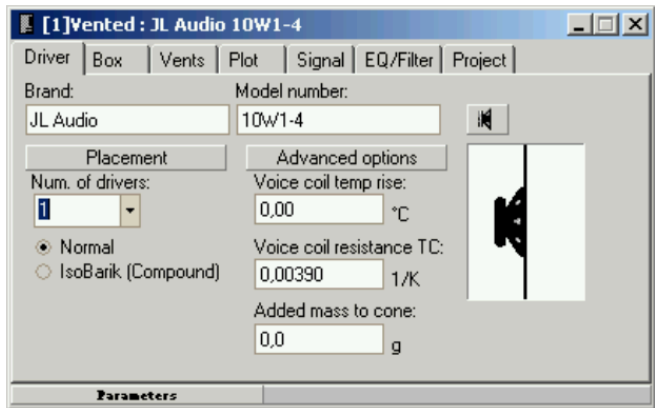
Designing your box (which is also referred to as your project since, hopefully, it will soon be a project in your garage) is accomplished by using the Project Window that is presented once you select a driver and the type of box you plan on putting this driver into from the database. Everything that you need to do to completely design your box (project) is done in the Project Window.

You can load as many drivers as you want (even multiple loads of the same driver if you wish), and each driver loaded will have it's own Project Window which "controls" its respective plot on the graph (each plot represents its corresponding Project Window). Each plot will also have its own color, which you can change by clicking on the color bar at the bottom of that plots' Project Window. The "plot", by the way, is the curved, green line (green in this example, you can make it any color you wish as described above) in the picture below. For demonstration purposes, go ahead and load the JL Audio 10W1-4 driver.



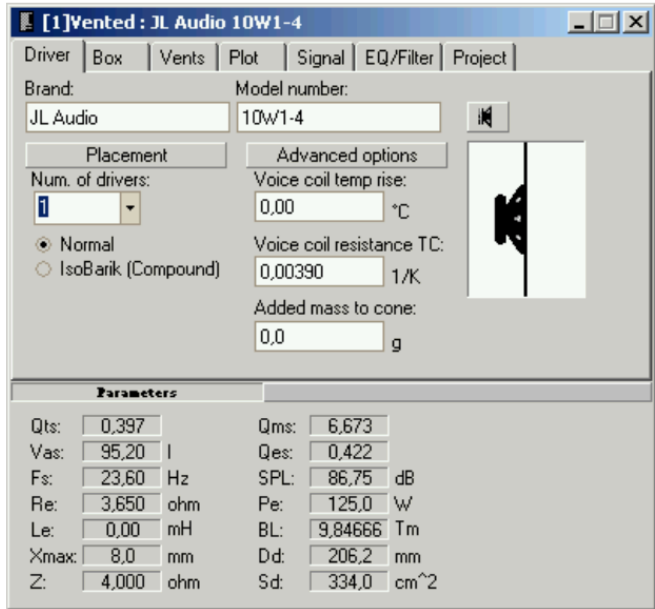
Again, everything you need to do to design your box is done in the ubiquitous Project Window! The Project Window has seven tabs (six if using a sealed box, since no Vents tab is necessary in that scenario, also when designing passive radiator system, the passive radiator has tab of its own), and each of these tabs are described below:

Driver Tab



Here you can view or change your driver's specs, as well as change how many drivers that you want to cram into your box, and (if more than one), how you plan to cram them in there...

Also, you can specify how much extra mass you place into each driver's cone. This way, you can make driver's resonance frequency into lower value. But nothing comes free. You lose sensitivity, and even worse, your driver Q-values will rise. That might be intolerable. Anyway, option is there in case you want to try it out. If you want to view or change your driver specs, simply click on the "Parameters" box, and a drop down window will appear showing you the specs of the driver:



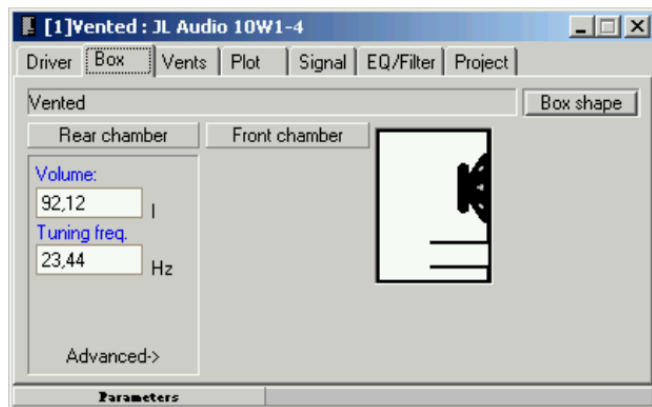
You can change the values and/or the units in which the values are displayed simply by clicking on them. This feature allows you to play "what if" with your driver (which can be very educational for learning what the different Thiele/Small specs control...try changing some of them and watching what affect it has on the plot), and is also ideal when using Driver Templates. Keep in mind, however, that any changes you make here are temporary, and will be set back to the default values once you quit WinISD. Note that you shouldn't make values ambiguous.

You probably have noticed the small driver icon besides of the model number entry field. If you want to do more detailed examination of driver data fields, you can "drag" the driver object into driver editor window icon in the "Editor"-button. You can do this by clicking and holding left mouse button in this icon and then dragging the icon into editor. This will open [Driver Editor](#).

In the Placement section of this window, you can choose how many drivers that you want to use in the same box, as well as whether they are to be in a "Standard" or "Isobarik Pair" configuration. Remember that the placement options you choose here are calculated using the box type (ie; sealed, vented, bandpass) that you chose when you initially loaded the driver. You cannot change the box type at this point. If you want to compare how the same driver performs in separate box types, then you must load the driver over again in a different box type. After all, you're gonna want to view separate plots for each type anyway, right?

Box Tab

This is the most powerful part of WinISD! While you can finish what you need to get done here within seconds, you will probably find yourself here tinkering for hours ...*read this section carefully!*



When this Tab is initially opened, the default Volume and Frequency data is displayed. Sometimes, the "default" box volume is too large to be practical, or maybe you are just building an addition to an existing speaker system, and a different plot is desirable. By manually changing the size of the enclosure (and/or port frequency for bass-reflex), changes in the output plot appear. These changes in the plot represent the real world performance of the completed box.

For bandpass boxes, there are two chambers, front and rear. Rear chamber is that what is normally in rear of the driver.

There are two ways to change the Volume of the Box (from here on out referred to as V_b) and the tuning Frequency of the Box (from here on out referred to as F_b (on vented boxes)). You can either do it manually by double clicking on the numeric values you wish to change, and then typing in new values (the graph changes with each keystroke entry, no need to hit the enter key), or you can do it "on the fly" (dynamically) with your mouse!

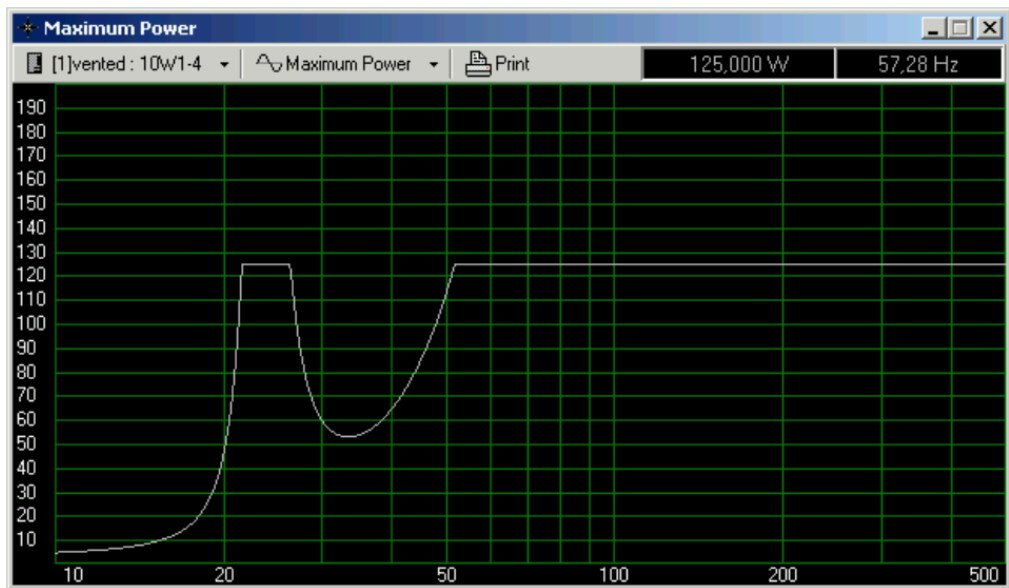
To change the values with your mouse:

- Ensure that you are in the Box Tab section of the Project Window.
- Position the cursor over the diagram of the speaker box (below the text which says "Box Shape").
- Click and HOLD the left mouse button while moving the mouse... Left/right movement of the mouse will change the V_b , and up/down movement will change F_b (on vented boxes). Notice that the plot curve changes automatically in unison with your mouse movements!
- Continue to move the mouse until the desired graph is displayed.

The power of this feature is that no longer do you have to manually type in changes that you hope will produce the desired plot! You simply "drag and plot" your desired curve and viola, the necessary box dimensions are already calculated and displayed for you! You can also "lock in" one parameter (either Box Volume or Tuning frequency) while changing the other. How? Click [here](#).

This, coupled with the ability to load and view the plots of multiple drivers at the same time, makes choosing the right driver for your application a breeze!

There are other concerns than just the frequency response. One of the most important ones is the mechanical power handling. That means, how much power our design will tolerate without running out of limits of maximum excursion. You can get estimate of this by looking at the "[Maximum Power](#)"-plot:



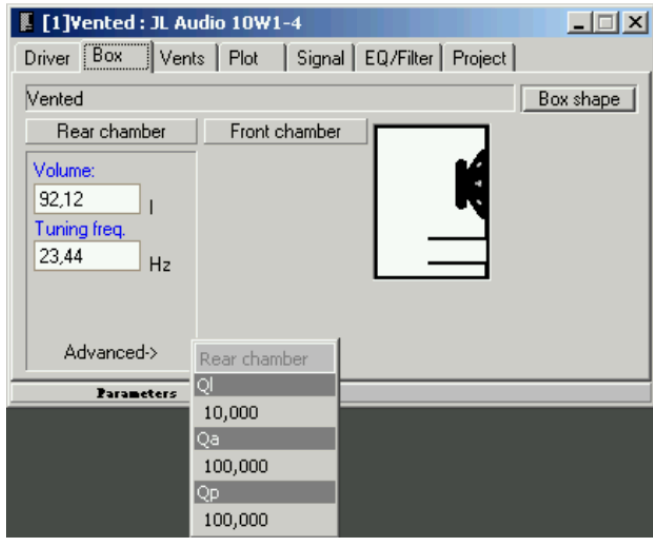
Graph shows us clearly the well known fact about ported boxes, that their power handling is severely weakened below tuning frequency. Best power handling is achieved around tuning frequency, where acoustic resonator represents heavy acoustic load to the driver.

Clicking on the Box Shape button activates the Box Shape calculator:

Under "advanced->", you can find controls for box losses. There, you can see what effect the each loss type will have on your box design (Smaller the value, greater the loss). There are three types of losses:

- **Ql**, leakage losses. These are produced by leaks in enclosure or in driver itself. Generally, this is most dominant loss type in vented boxes. Typical value is 5-20. This is quite impossible to predict before building the actual box. For reasonable quality box, WinISD pro uses Ql of 10 by default.
- **Qa**, absorption losses. These are produced by losses in enclosure. Any stuffing will increase the absorption loss. With no stuffing inside the box, 100 is typical. Heavily stuffed enclosure has this about 3-5.
- **Qp**, port losses. These are produced by port. Port has some resistance (air doesn't actually flow through the port without some friction). Actually, by setting this into very small value, you can turn vented box into closed one!

When you click on the "Advanced->", a popup menu opens, where you can see what current loss parameters are:



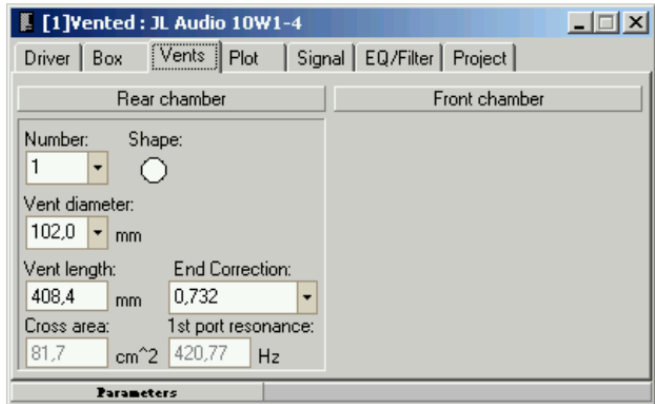
When you choose a particular loss type from this popup-menu, a small window will appear:



In this case, I choose "Q!". Here, you can enter new value for loss or you can use similar drag method, by clicking on the white square beside input box. Again, the graph will update in real time.

Vents Tab

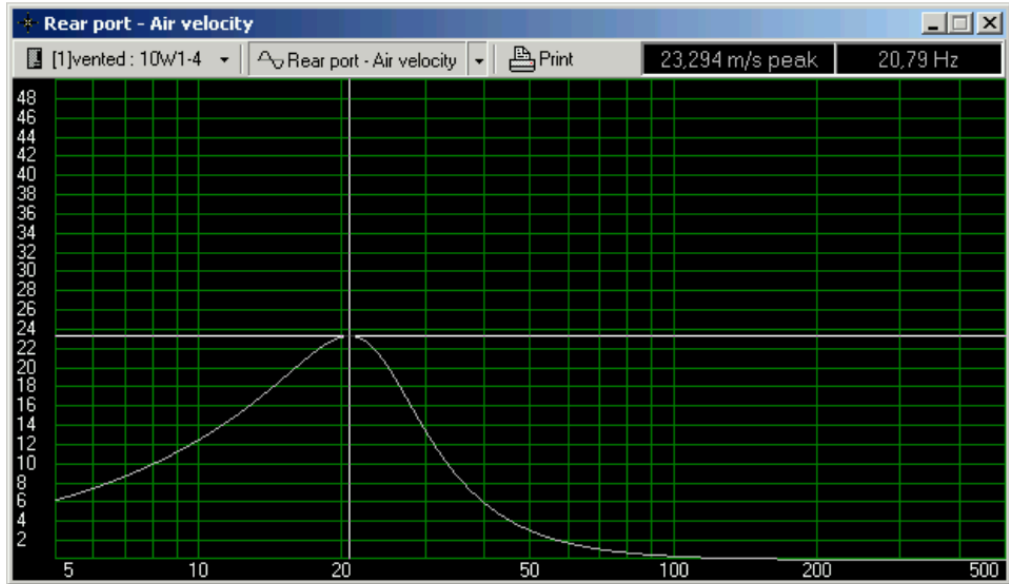
Clicking on the Vents Tab displays the vent (port) information.



Here, you can decide how many and of what shape you want your vents to be on your box. The ports, of course, serve to "tune" your box to the frequency that you have chosen while you were designing your box in the Plot Window. The size and -3dB (cutoff) frequency of your box determine the size of the ports.

1st port resonance shows frequency of first "organ-pipe" resonance. This helps you to estimate where port becomes undesired radiator.

Very important here is to make your ports large enough, so that peak air velocity won't exceed 5% of sound velocity in air. That is, about 17 m/s assuming normal environmental conditions. You can check it by using "[Rear port - Air velocity](#)" graph, by first setting simulation power to anticipated maximum power, see "signal"-tab. In this case, P_e is 125 Watts.



Here, the velocity peaks at 20,8 Hz. Peak air velocity is about 23,3 m/s, which is a bit high value. This might or might not be a problem, depending on your program material; If you don't expect very powerful sounds at vicinity of this frequency, you shouldn't have any problems. However, if you want to play safe, use larger port or several ports.

If your port velocity is too high, the only way to lower it is to either increase the number of or the diameter of the ports. Of course, here comes that dreaded "trade-off" word again! If you make your port(s) bigger in diameter, they will be able to pass more air with less effort. However, in doing this, you have lowered the "loading" of the speaker(s) in the box. This, of course, changes the -3dB point of the box. The only way to get back to your desired -3dB point is to "load" the speaker(s) again. How? Well, you gotta make the new, larger port(s) longer! This forces the speaker(s) in the box to work harder to move the air into and out of the box, and brings your -3dB back to where it should be. THE END OF YOUR PORT(S) MUST BE AT LEAST ITS DIAMETER AWAY FROM ANY INTERIOR WALL OF YOUR ENCLOSURE.

"Ok, so I make the ports larger, big deal?" you say. Well, if you're building large speakers for the home, then that's probably not going to be much of a problem for you. However, if you're building small, car sized speaker boxes, then the increased length of the ports is indeed a big problem, since you will need more port length than your box has room for!

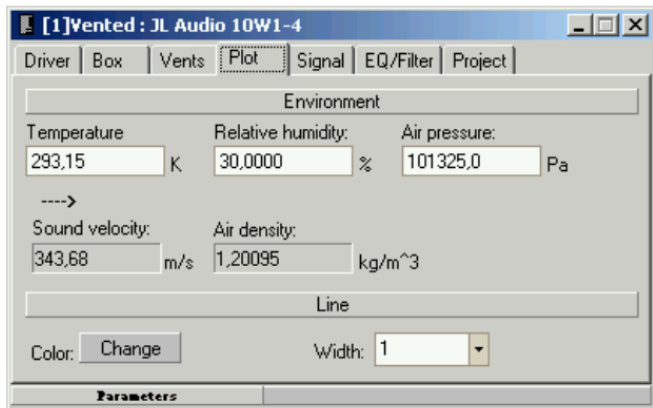
There are currently three ways to combat this problem:

1. *Use flexible tubing.* Flexible tubing is designed to be able to twist and turn inside the enclosure, so that if your ports need to be 14" long, and the longest side of your enclosure is 12", you can just bend and twist the tubing in there to get it done. The bad news here is that this approach takes up an awful lot of interior space in your box, which you'd better account for, else your box will sound like crap. Also, that flexible tubing isn't very flexible. It has a turn radius similar to a school bus.
2. *Use rectangular ports.* Rectangular ports give you "more for your money" as far as length goes. What I mean by this is that using rectangular ports enable you to squeeze more "port power" into a smaller box. The reason for this is that rectangular ports can follow the walls of the enclosure (not unlike a maze) in order to acheive the desired length. The bad news here is that rectangular ports must be built into the box, and therefore require much more skill and equipment to accomplish. As with flex tubing, rectangular ports consume a lot of internal space, and so the entire design must be re-done to allow for the area displaced by the ports.
3. *Run "external" ports.* Yes, you can run your ports outside of the box (the box doesn't care). This allows for a minimal box size as compared to traditional ports. The bad news here is that your box will resemble a space-age life support system, and people will call you eccentric.

If none of the above options appeal to you, then you must go back to the drawing board and make the sacrifice somewhere else. Remember, a small box tuned to a low frequency will demand a large port design. It's that damned "trade-off" again. Yeah, you can have a small box deliver big sound, but, as you now see, it's requires more thought and more work. But, the good news is, it CAN be done.

You shouldn't try to make too long ports, because they must be very much shorter than wavelength of sound to be reproduced by the system. About 1/20-1/10 is safe factor. Otherwise, they start to behave like organ pipe, which is not desirable. If this proves impossible, then viable solution might be using passive radiator instead of the port. However, it has its own peculiarities, which you can see, when you simulate such a system.

Plot Tab



Here, you can set values of several environment variables. This is because sound velocity is dependent on environment values. Default values are ok for casual user. If you try changing the temperature, you'll notice that it affects the port length. As you can see, the effect can be quite great! This is one of the difficulties, what most box-building programs won't deal with at all. Calculated sound velocity is shown also. Calculation of sound velocity and air density is derived from Claus Futtrup's excellent documentation of **Driver Parameter Calculator (DPC)**.

You can also change line color and width.

Signal Tab

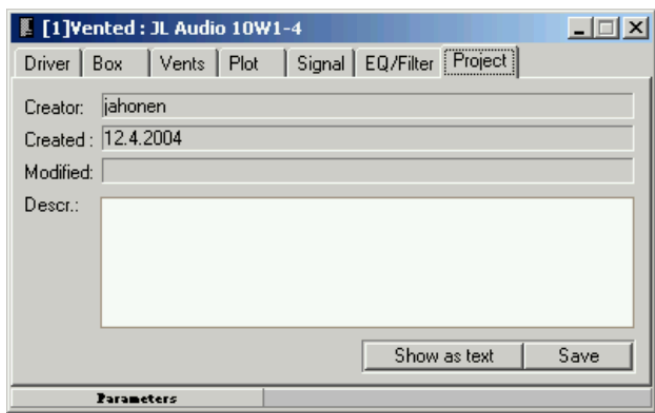


Signal source has two properties, Power and Series resistance. Term "power" should more correctly be voltage. This term "power" comes from definition by Richard Small, who defined the input power to be $P=E_g^2/R_e$ (this is to simplify things, because impedance varies greatly depending on the frequency and the box), where E_g is RMS output voltage of your amplifier, and R_e is DC resistance of voice coll. Actually, driver's impedance will vary greatly (see impedance graph), depending on the frequency, so this is generally not the power which is taken from the amplifier.

Series resistance is source resistance seen by the driver. This includes your amplifier's output impedance, crossover coils, connectors, interconnect wires. Raising this value will effectively rise the Qes. Default value is 0.1 ohms, which is reasonable default. If you experiment, you'll find out a quite large change in system behaviour.

Distance is distance from imaginary speaker in half-space, assuming far field on-axis conditions. It is not true on very small distances, although some people have quite succesfully simulated in-car SPL levels using small values.

Project Tab



The Project tab allows you to save your Project, complete with all of the changes that you made, to disk for later retrieval.

Clicking in the Descr: field opens a dialog in which you can type some text to describe the project.

Clicking on the Show as text button brings up the Project as Text window:

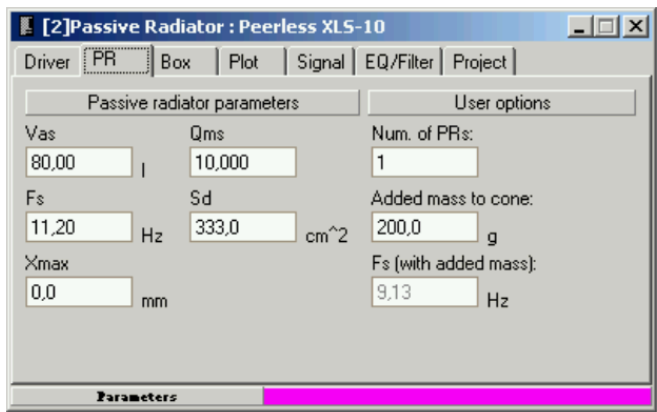
When you're done, you can either save it in the default directory (WinISD), or you can create a new directory (recommended).

Next time you want to work with this saved project, you can click on File, then Open Project, and then select your project from the folder that you saved it in.

Passive Radiator Tab

This tab is used only with passive radiator systems. For demonstrating the thing, we must create project for passive radiator system.

Basically, "PR"-tab shows relevant parameters of your passive radiator unit. Parameters are same than any ordinary driver. You can in fact use normal drive unit instead of passive radiator, just remember to leave it's terminals unconnected. Like active drivers, you can place several passive radiator units in your box design. There is also possibility to add mass to passive radiator cone. If there are several PR units, mass is per unit mass. WinISD shows below unit's new resonance frequency with added mass.



Filter/EQ Tab



This concludes the Designing Your Box Help File section. Be sure to check out all of the articles sections! -JJ

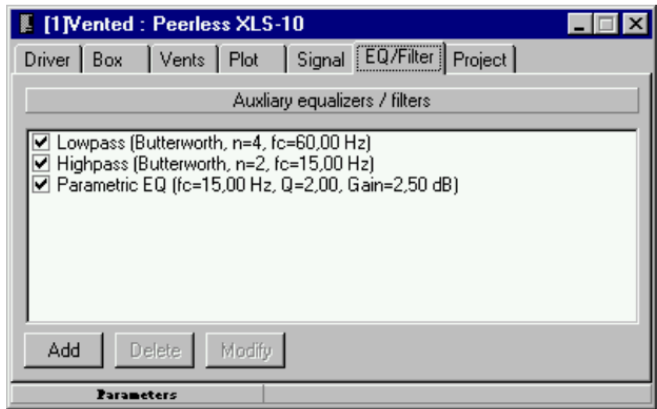
This page originally written by JJ Richard (JJRichard@linearteam.org) and revised by Janne Ahonen (janne@linearteam.org)

Filter/equalizer behavioral simulator

WinISD pro supports behavioral simulation of various filter types, such as:

- Butterworth low- and highpass, orders 1 thru 10.
- Linkwitz-Riley low- and highpass (fourth order only).
- Second order high- or lowpass section, with user specifier Q and fc.
- Bessel low- and highpass, orders 1 thru 10.
- Allpass.
- Linkwitz transform.
- Parametric EQ.
- Second order highpass with user specified peak gain and peak frequency.
- Static Gain.

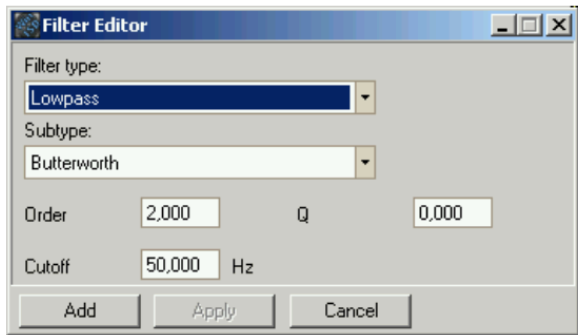
The behavioral simulation means, that WinISD simulates filter as "black box", it does not consider what black contains internally. Filter can be passive or active, as long it does what is simulated.



Any combination of cascaded filters can be simulated. Filters are connected logically in series. Individual filters can be turned on and off by checking/unchecking the checkbox in the listbox. Graphs are updated immediately, so effect can be investigated.

Filter system is logically located at electrical side. 0 dB gain at filter chain means that voltage at driver terminal is equal that is specified at "signal"-tab. If filter/EQ chain gain is 6 dB at particular frequency, then voltage at that frequency is twice that is specified, and so on.

Lowpass filters



The following lowpass filter subtypes can be simulated:

- Butterworth (orders 1 thru 10)
- Linkwitz-Riley (4th order only)
- Bessel
- Second order section (SOS) with user specified fc and Q.

Explanation of the filter subtypes

Butterworth

Butterworth is probably most common filter type. It is so called "maximally flat" filter. This means that amplitude response stays practically flat until fc frequency is reached, where it is -3 dB down. Response then falls rapidly according the filter order.

Linkwitz-Riley

Linkwitz-Riley is filter type, where two second order butterworth filters are cascaded as one fourth order filter. This means that frequency response magnitude is -6 dB down at fc. It is different from fourth order butterworth, where magnitude response is -3 dB down. Transient response is better than butterworth, but still worse than bessel.

Bessel

Bessel filter is fundamentally different from two above. Whereas butterworth is built on amplitude response approximation, the Bessel filter is maximally flat group delay approximation filter. fc is frequency, where the phase is half of the maximum phase shift (90 degrees times the filter order). This filter exhibits best transient properties of analog filters, but roll-off starts rather slowly.

Second order section

This filter type is for advanced users. Second order section can be considered as "typeless" second order lowpass. Parameters are fc and Q. fc determines the frequency, where the filter gain is Q or 20*log(Q) dB. This can be used to compose new "all-pole"-filters, by adding several sections in cascade.

Highpass filters a.k.a. Subsonic

Highpass type contains same subtypes as lowpass filter. See explanation above. Only difference is that cut-off is below fc, not above, as is logical.

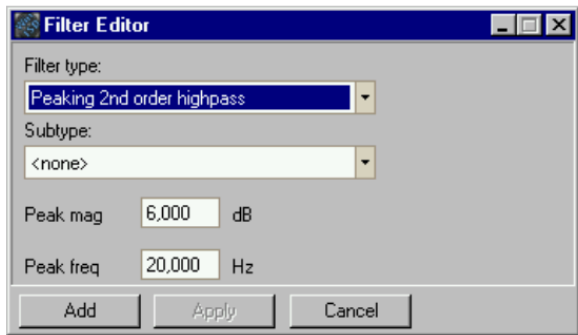
Lowpass/highpass filter parameters

Order determines how rapidly the response magnitude decreases after cut-off point. Greater the order, steeper the roll-off. This has it's price, however. Faster you make it roll-off, worse the group delay behaviour. 1st order filter has roll-off of 6 dB per octave, 2nd order filter 12 dB per octave, and so on. Realization of high-order filters is very difficult to achieve, due to high tolerance requirements. Generally, anything over 4th order should be considered very carefully.

fc Cut-off frequency or critical frequency. For butterworth class of filters, this is the -3 dB point.

Q is only used with Second order section-type. This makes possible for the user to experiment with filters with peaky response (however, this is seldom done with lowpass filters). Q is the quality factor of the filter. It determines the shape of frequency response near the fc frequency. One physical meaning for Q is that gain at frequency fc is Q (in linear scale, not dB).

Peaking second order highpass



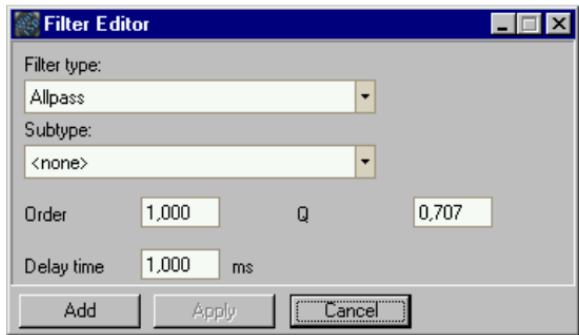
This type of filter is commonly used as subwoofer equalizer. It suits quite well to ported or passive radiator type boxes. Because tuning frequency of the box is quite near the roll-off, response can be boosted several dB's before cut-off, extending the frequency response. After the peaking, response starts to roll off quickly. This is convenient, since it serves as an subsonic filter also. However, all this comes with a price tag: it introduces quite high amount of delay distortion depending of the amount of

peaking.

parameters

Peak freq is frequency of maximum equalization.
Peak mag is amount of equalization in dB you want to apply at peak freq.

Allpass



This is so called phase correction filter. It alters only phase, gain is unity at all frequencies. Only first and second order allpass filters are implemented.

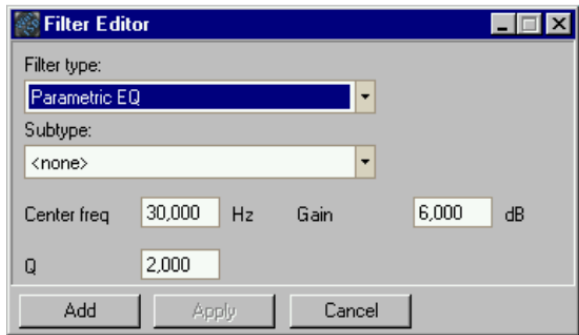
allpass filter parameters

Order determines how fast the delay will reduce, after cut-off point. And of course, the practical implementation is quite different.

Q or the quality factor. This is relevant only with second order all-pass. Higher the Q, higher the peak in group delay.

Delay time is desired signal delay time. Delay approaches this value asymptotically when frequency approaches zero. Higher you set this value, earlier it rolls off.

Parametric EQ



This type of filter is implemented in commonly available equalizers. They might come as freely adjustable, where center frequency, boost/cut and quality factor are all adjustable (they are called as parametric equalizer), or partially adjustable, where one parameter is fixed (semi-parametric equalizer) or only boost/cut amount is adjustable (they are called graphic equalizers). You can boost or cut desired band.

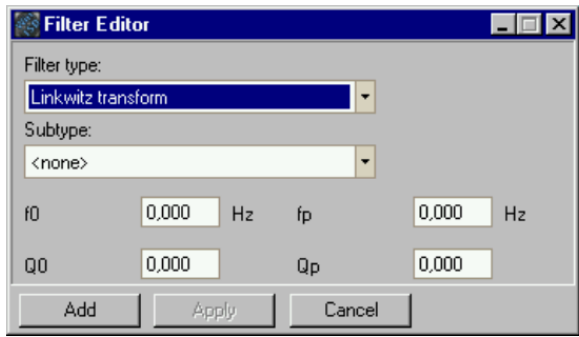
parametric EQ parameters

Center freq is frequency of maximum boost or cut.

Q controls bandwidth of the boost/cut. Higher the Q, narrower the equalization is.

Gain adjusts amount of the boost/cut. Use negative values for cut, positive for boost.

Linkwitz transform



This is an elegant way to equalize closed box subwoofer. The good thing about this is that it introduces no excess phase shift, which would be inevitable introduced if for example, parametric equalizer would be used. In practice, Linkwitz transform makes closed box look like the specified parameters. So called S-plane representation would be nice way to show how this thing works, but it is too difficult concept to grasp without engineering background.

Linkwitz transform parameters

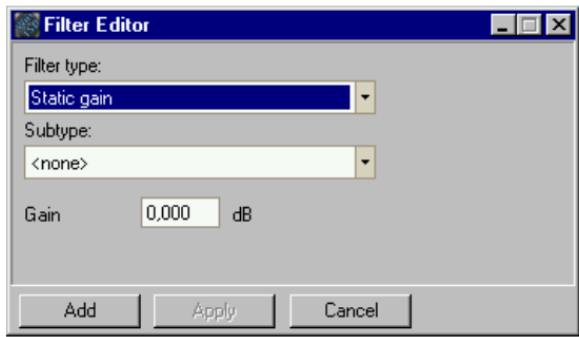
f0 is fsc of original closed box system. WinISD automagically fills this field, if adding to closed box system.

Q0 is Qtc of original closed box system. This is also filled automagically, like f0.

fp is fsc of the transformed system. This can be rather freely specified, of course one should consider the physical limits.

Qp is Qtc of the transformed system. This is also up to user. General guideline is to keep this below 0.707, or otherwise peaking will occur.

Static gain



this "filter" just adds described amount of dB to the eq/filter response curve. It can be used to normalize the response, to user preference. For example, there has been a debate whether the linkwitz transform should be normalized at high or low frequencies. With this, it could be either.

Graphs related to the filter simulator

There are three purely EQ/Filter-related graphs in WinISD:

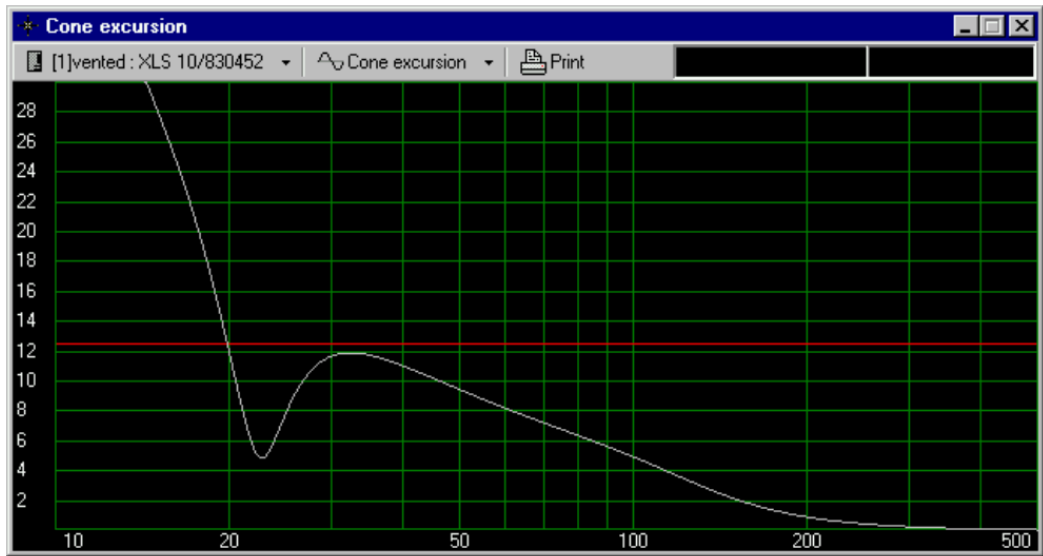
- EQ/Filter transfer function magnitude
- EQ/Filter transfer function phase
- EQ/Filter group delay

These are same curves than system curves, but calculated just for the filter part.

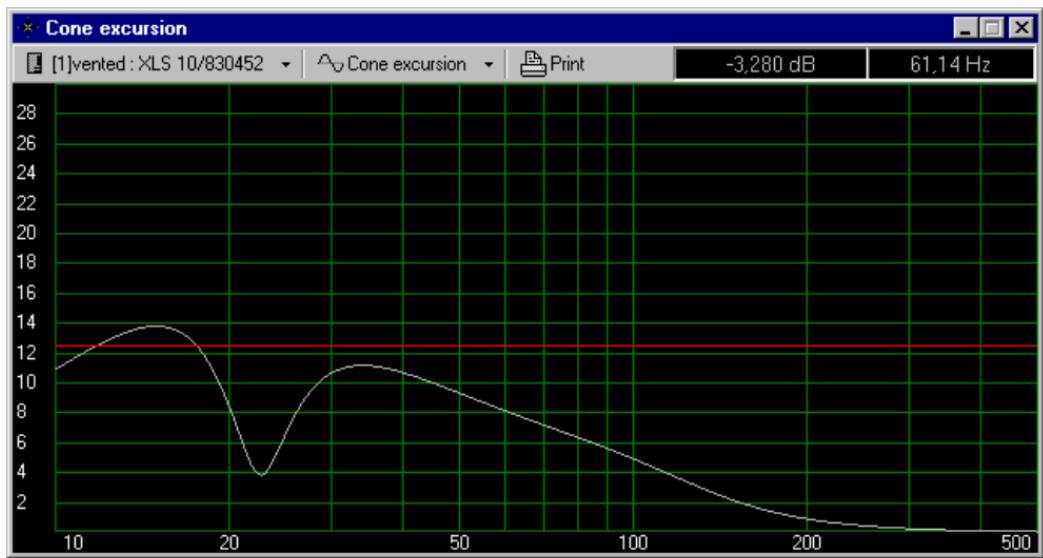
Filter simulator examples

1. Adding a subsonic filter

A subsonic filter is quite often required in ported systems, due to the fact that driver becomes unloaded below tuning frequency. This is illustrated in graph below. Tuning frequency is 22.65 Hz. Below 20 Hz, cone excursion starts to rise to the sky.



If the program material should contain low frequency signals below 20 Hz, then driver is driven out of it's linear range and it will produce distorted output, or might be even damaged. Let's add 20 Hz second order subsonic filter, go to the EQ/Filter tab, and click on "Add"-button. Choose "filter type" as highpass and subtype as butterworth. Enter order as 2 and cutoff as 20 Hz. Q value does not matter in our case. Then click on "Add". The cone excursion graph should be much better now:



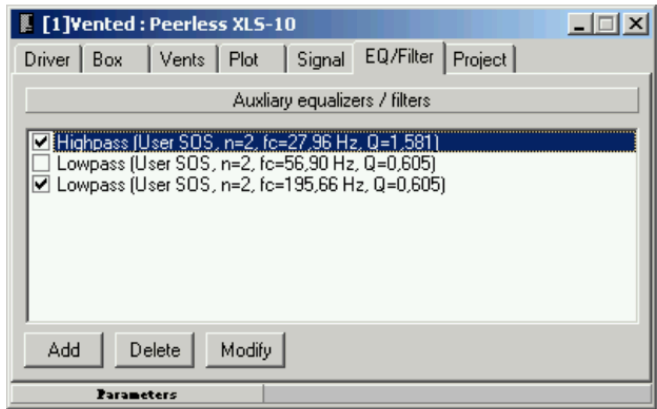
2. Simulating a complete filter

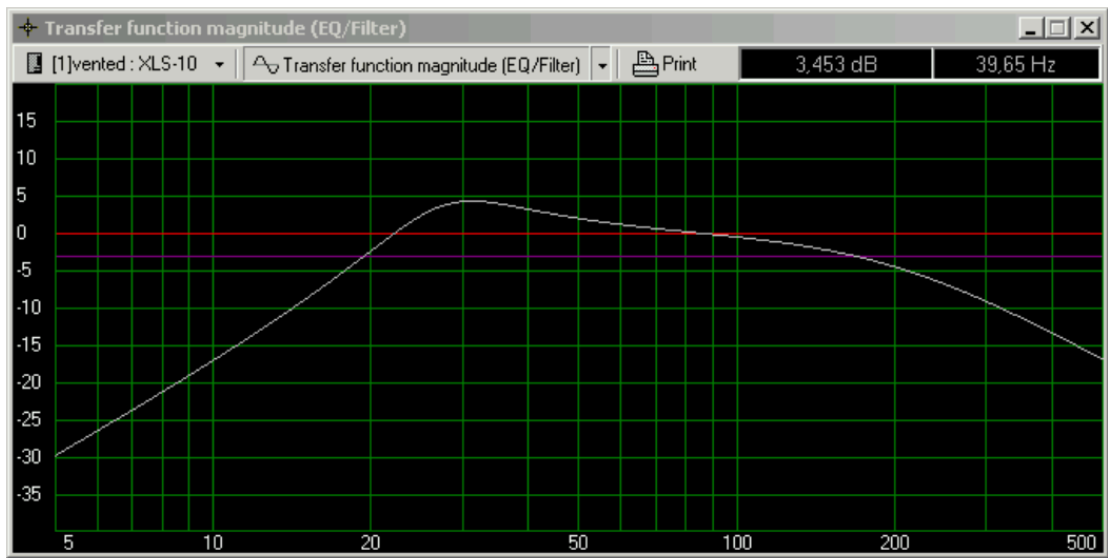
One example might be a subwoofer plate amplifier, which is sold under "Simex"-brand here in finland (type BA-480). Same amplifier is sold by Parts Express (300-794). It's subsonic filter has f_c of 27.96 Hz and Q value of 1.5811. Lowpass filter's f_c is adjustable, from 56.90 Hz to 195.66 Hz, with fixed Q value of 0.60486.

filter	f_c /Hz	Q
subsonic	27.96	1.5811
lowpass (minimum)	56.90	0.60486
lowpass (maximum)	195.66	0.60486

We have to use Special second order section-filter, because of the weird Q-value.

Adding these filters to the filter chain and disabling the lower lowpass results following response curve:

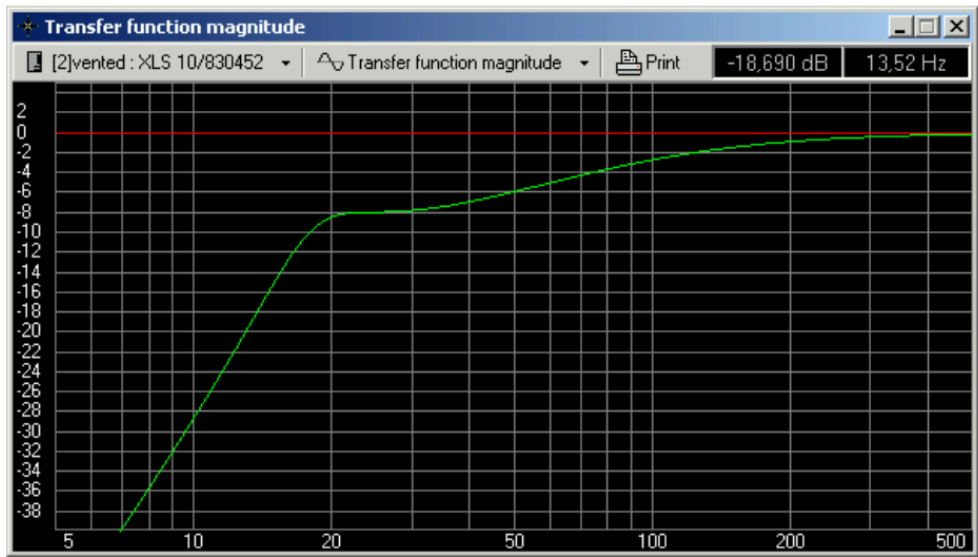




This page written by Janne Ahonen (janne@linearteam.org).

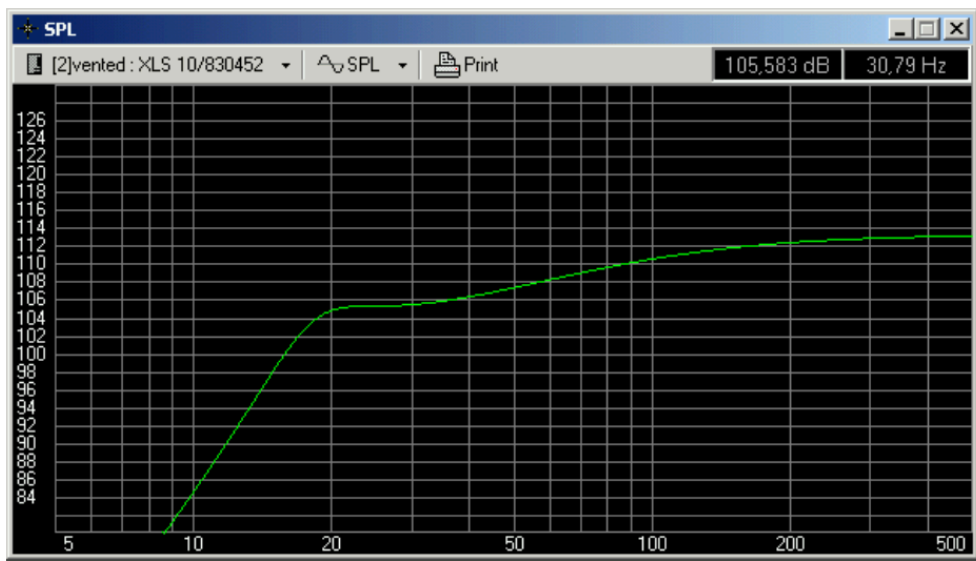
What those graphs tell you?

Transfer function magnitude



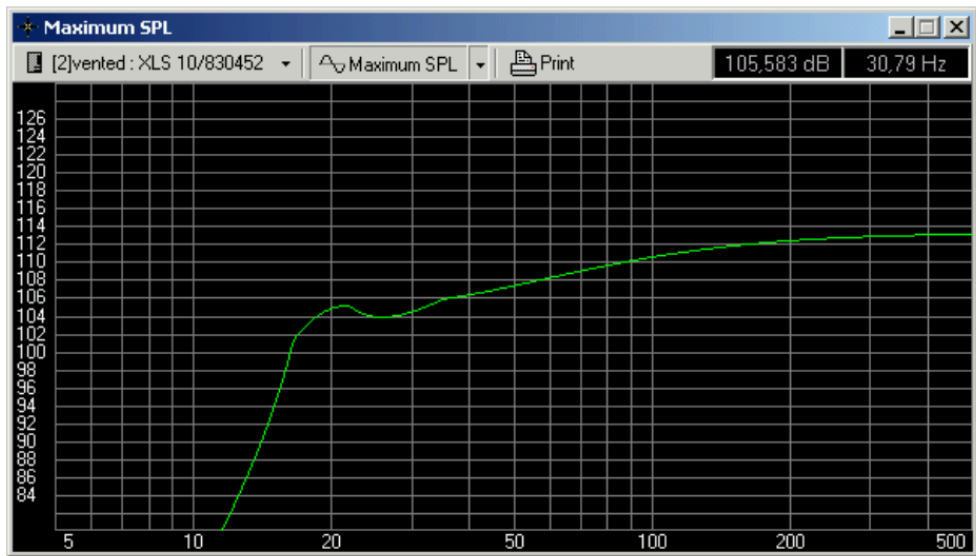
This graph shows gain in dB related to driver's limit efficiency n0. Which it theoretically reaches at infinite frequency. This is the basic graph, which is needed most.

SPL



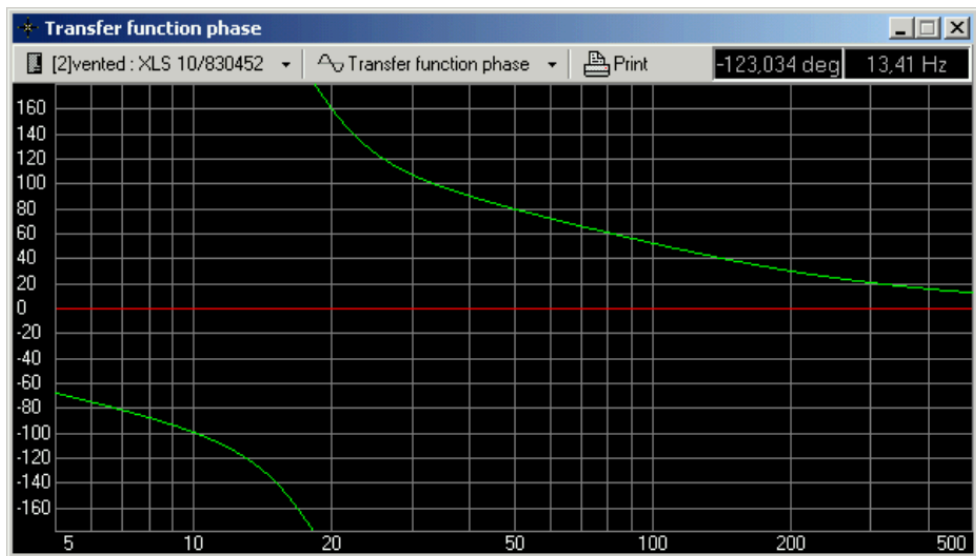
SPL graph shows Sound Pressure Level at specified distance at specified powerlevel, radiated to half(2pi) space. To get the full-space value, subtract 6 dB from the reading. Because driver's efficiency is related to ambient conditions, changing for example the project's temperature, will change the calculated SPL level.

Maximum SPL

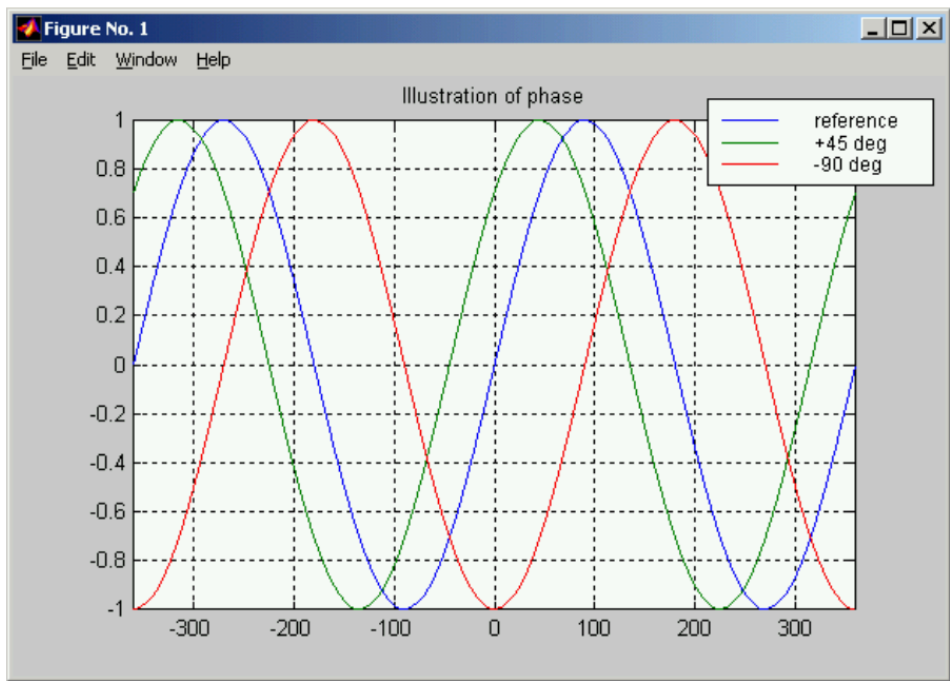


Maximum SPL graph is a combination of SPL and maximum power graphs. It tells how loud particular design is able to play, considering Xmax and Pe restrictions. If calculated power to reach Xmax is greater than Pe, then Pe is taken as input power to the driver. With this, you can easily visualize potential of particular design. Please note that it doesn't apply to room or in-car environment, what could change things quite a bit. It is useful as an comparison tool, e.g. comparing same driver in closed or vented box. See also [Maximum Power](#).

Transfer function phase

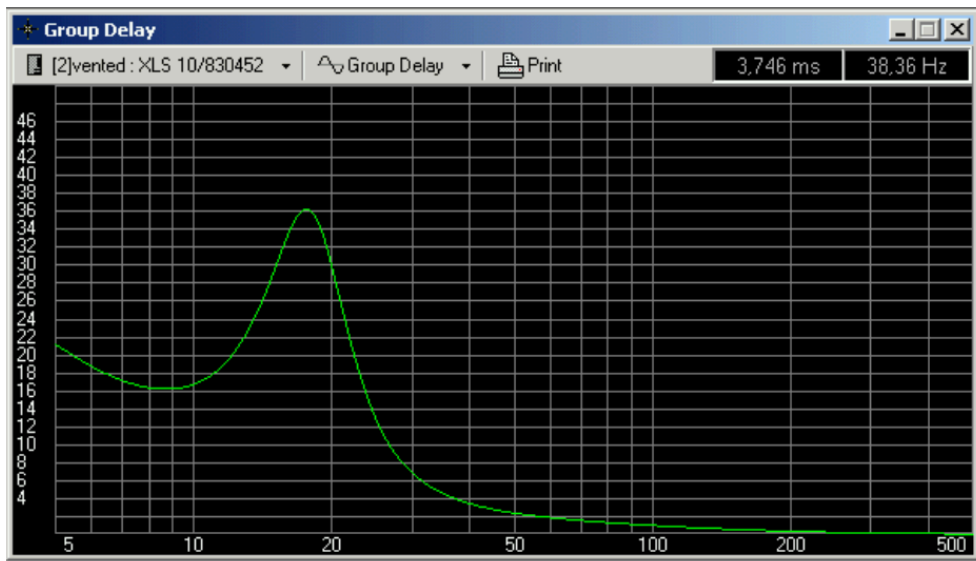


Phase shows the phase difference between the electrical input signal and acoustic output signal. See following illustration:



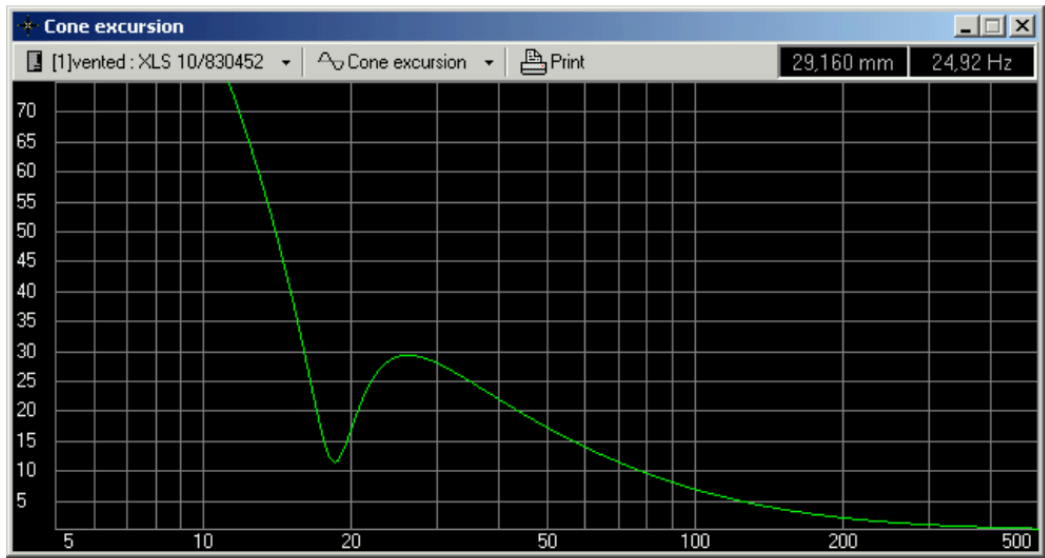
Here, the blue plot is the zero-phase signal, the reference. You always need a reference to define the phase shift. Now, if we add phase shift of +45 degrees, we get the graph plotted with green color. So it is kind of "ahead" of the input signal. EE people say that it has "*phase lead of 45 degrees*". On the other hand, if we make phase shift negative, we get that red plot. There, the phase shift is -90 degrees. So it is said that there is a "*phase lag of 90 degrees*". Now, you'll probably wondering how this relates to phase plot WinISD is showing to you. You can take the input signal which your amplifier will feed into your speaker to be that blue reference. Phase graph tells you how much phase shift in degrees there will be between outputted acoustic signal and input signal.

Group delay



Group delay describes how long spectrally infinitesimally narrow sine burst signal takes to travel through speaker's acoustic "filter". The flatter the group delay, the better. The value itself is not a concern, other that if delay becomes too large, then it might be difficult to match the design to other speaker's which might have less group delay. Large groupdelay variations usually means large ringing when transient signals are applied to the system. Mathematically it is the derivative of negative phase versus frequency in radians/sec. So when phase changes linearly versus frequency, then the group delay has a constant value. If that seems to hard to think, think it as follows: Take a number of people, where each individual represents a particular frequency. Now, send them simultaneously running to specific target. If they all reach the target at same time, then group delay is flat. They all take same time to reach the goal. Now, if some of those are in worse shape than others, they'll arrive later than others. Now, the group delay is not constant, and different frequencies will arrive at different times. That's group delay.If you are wondering why the group delay of vented box is very different from WinISD version 0.43 and below, the reason is that it calculated it wrong. Current versions produce correct results.

Cone Excursion (driver/passive radiator)



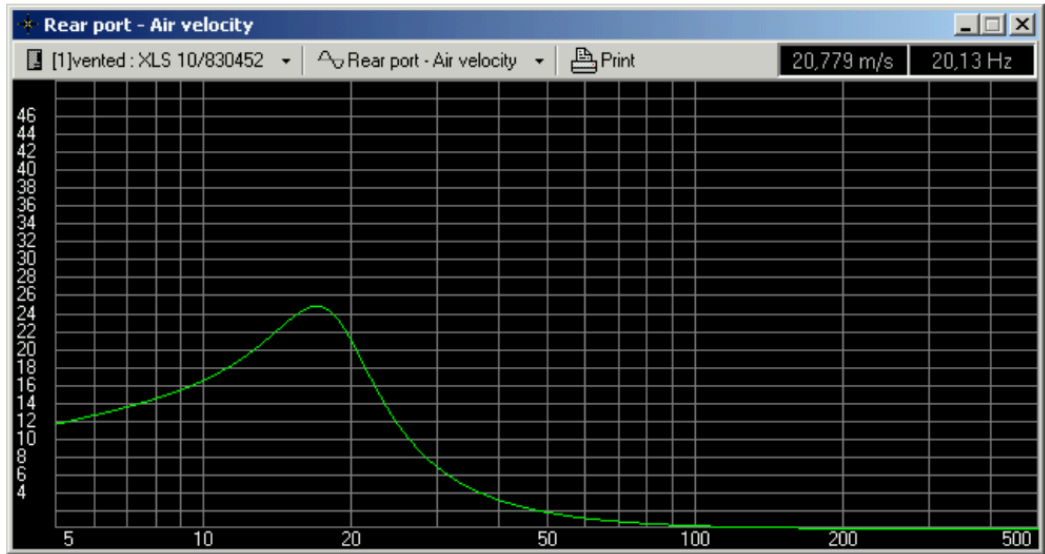
Cone excursion shows how much driver cone moves with sinusoidal excitation at chosen powerlevel. The powerlevel is controlled in "plot"-tab. The power applied can be related to excitation voltage with following relation: $E_g = \sqrt{P \cdot R_e}$, or $P = E_g^2 / R_e$ where E_g is the RMS voltage applied to driver's terminals, P is the input power in watts and R_e is the DC resistance of the voice coil. Please note that there is few different ways to express this value. WinISD can be configured to show RMS, Peak, Peak-to-peak (p-p)

values of the cone excursion. RMS value is defined just as RMS value of any sine waveform. Peak value is the difference between zero and the maximum value of sinusoidal waveform. Peak-to-peak is twice the peak value, i.e. difference between minimum and maximum point of waveform. The peak value is perhaps a most practical expression, because driver parameter Xmax indicates how much cone can be deflected from it's rest position linearly, in either direction.

If you want to maximize power handling of any box, then adjust the box parameters so that cone excursion is kept at minimum value possible. Of course the transfer function magnitude graph should be taken into consideration also. In closed box, the minimum excursion is obtained, when enclosure is as small as possible. Same basically applies to vented box, but there is a local minimum at port tuning frequency.

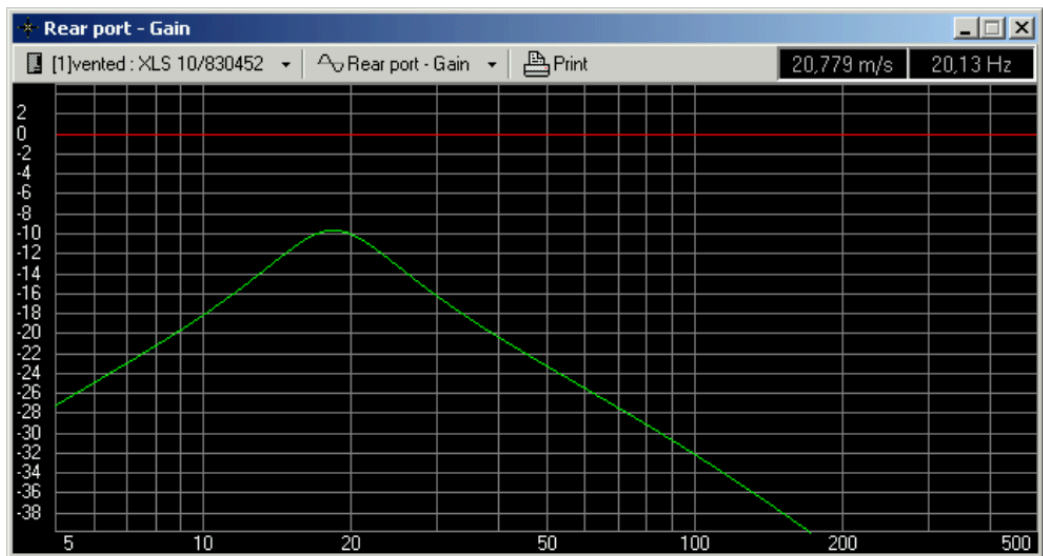
When comparing graphs between programs, please note that many programs give the RMS excursion which is "wrong", in my opinion. I have seen some programs, where the calculated excursion is RMS value, and limit is shown as peak. That gives over-optimistic power handling impression. Please also note that this graph doesn't take nonlinearities into consideration. But it let's you see when the nonlinearities are becoming too great.

Air velocity - front/rear port



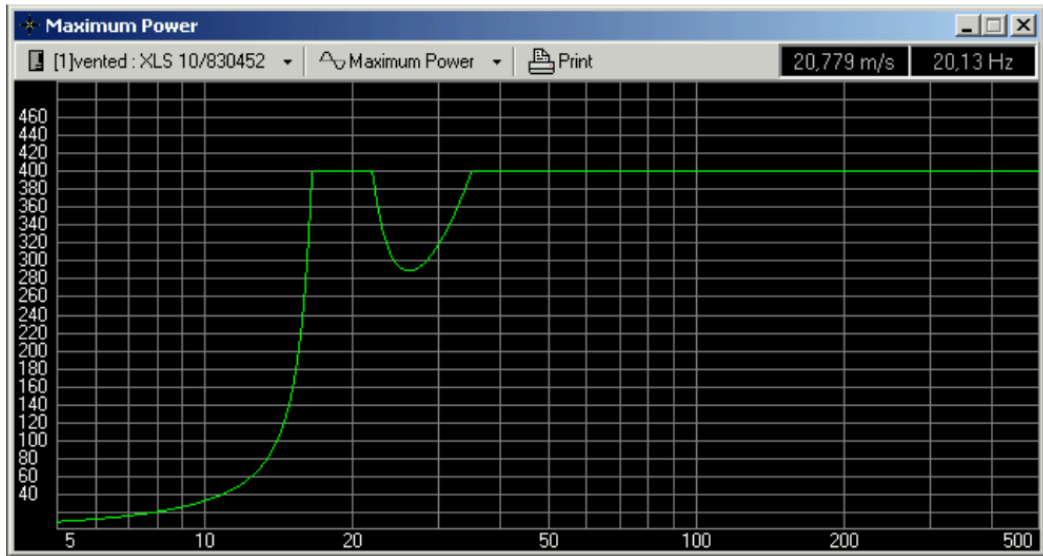
Air velocity graph shows how fast air mass travels in port. In order to keep chuffing noise low, you should limit the peak velocity at 5% of velocity of sound, or about 17 m/s. Like the cone excursion graph, the desired powerlevel is set also by same watts setting. Note that if air velocity peaks exceeds previously mentioned level. This graph can also be configured to show RMS, peak or peak-to-peak value. See the cone excursion graph explanation for RMS, peak and peak-to-peak values.

Gain - front/rear port



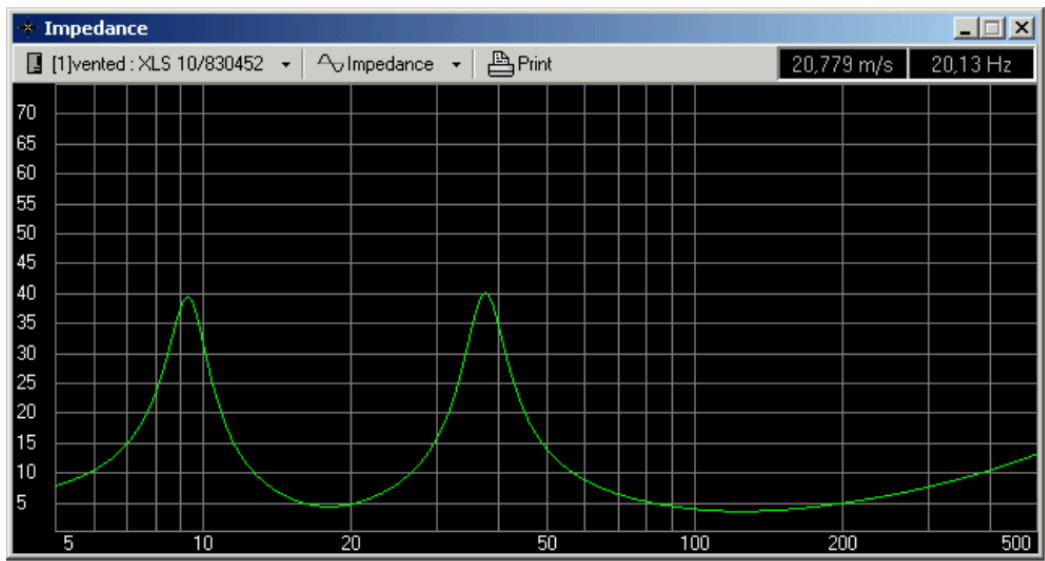
This graph shows n_0 (the driver reference efficiency) relative magnitude output of the front or rear port. In conventional vented box, this graph shows relative sound pressure contribution of the vent.

Maximum power



Maximum power indicates, what is the maximum powerhandling of the speaker at each frequency. It shows either the P_e , which is maximum electrical input power based on thermal power handling. Or if it is less, then it shows, how much power is needed to drive the cone to maximum excursion X_{max} . Again, you can obtain the required drive voltage by equation $E_g = \sqrt{P \cdot R_e}$, where the E_g is RMS terminal input voltage of the driver, P is the input power and R_e is DC resistance of the driver voice coil. See also [MaxSPL](#).

Impedance/impedance phase



Impedance graph shows, what kind of impedance load the amplifier will see. The impedance shown is the impedance of each driver, if there is many of them or there is an isobarik configuration. In LspCAD terms, drivers have "Separate Sources". The lower the reading, then higher the load on the amplifier. Impedance graphs shows actually the modulus of the complex impedance and phase angle (impedance phase) graph shows whether the load is resistive (phase about 0 degrees), inductive (phase positive) or capacitive (phase negative). Reactive load doesn't actually dissipate any power, but instead of dissipating it, it returns it to driving amplifier. In classic linear amplifier, this power is wasted. D-class amplifiers utilize this and return this energy to the power supply. High capacitive load is difficult for any feedbacked amplifier, because it eats the phase margin of the amplifier and may cause it to begin oscillating. This impedance gives only steady-state impedance for sinusoidal signal. For more complex input signal, the current drawn from the amplifier may be greater than calculated with this graph, because of energy storage of the impedance, although this may be very rare with normal audio signals. There is an AES paper about this written by Matti Ojala and Pertti Huttunen.

Entering your own drivers

If you want to create your own drivers, open the New Project window (CTRL-N is a shortcut), click on the "New" button next to the down arrow of the drop down on the New Project window. This opens up the Driver Editor window:

Driver editor

GeneralParametersAdvanced parametersDimensions

Manufacturer

BrandModel:

Data provided by:Date added:

jahonen

12.4.2004

Comment:

Entered

Calculated

Not available

Save

Load

Clear

☒ Auto calculate unknowns

Close

Again, driver editor is split into several tabs, which we will discuss now. Let's begin with first "General"-Tab.

Most of the parameters in "General"-tab are self-explanatory, but perhaps short explanation is in place anyway.

- Manufacturer is the name of the company who actually made the physical driver, e.g. name of the OEM factory.
- Brand is the marketing name of the driver.
- Model is type marking on the driver.
- Data provided by field is by default filled by user name.
- For comment field, you can enter your own findings about particular driver.

At bottom of the driver editor window, there is a "Auto calculate unknowns" option. As you'll later notice, it is very great help when entering parameters. In fact, it is recommended that you enter as few parameters as possible, because then Driver editor will ensure, that your data is in perfect accordance with Thiele/Small-model.

Driver editor

GeneralParametersAdvanced parametersDimensions

Thiele/Small parameters

Qes0,000Qms0,000Qts0,000Fs0,00Hz

Vas0,00l

Electro-Mechanical parameters

Mms0,0gCms0,0um/fRms0,00000H/mRe0,000ohm

BL0,00000TmDd0,0mmLe0,00mHsd0,0cm^2

fLe0,00HzKLe0,000000H*sqrt(Hz)

Large-Signal parameters

Xmax0,0mm peakHc0,0mmHg0,0mmVd0cm^3

Xlim0,0mmPe0,0W

Miscellaneous parameters

no0,0000%Znom0,000ohmUSPL0,00dB SPL0,00dB

Voicecoils1

ConnectionParallel

EnteredCalculatedNot available

SaveLoadClear

☒ Auto calculate unknowns

Close

In Parameters tab, you enter the [Thiele/Small](#) Parameters of the driver that you wish to add to your database. If you wish to change the mathematical units in which your parameters are given in, simply click on the unit and WinISD will toggle through the available units and make the conversion for you. For instance, if you put a speaker diameter of 8.0 in., and you'd rather have that value in cm, mm, etc., simply click on the value (in. in this case), and watch as WinISD converts it over for you. The area and distance systems are used for Ports and 1w/1m (SPL) measurements.

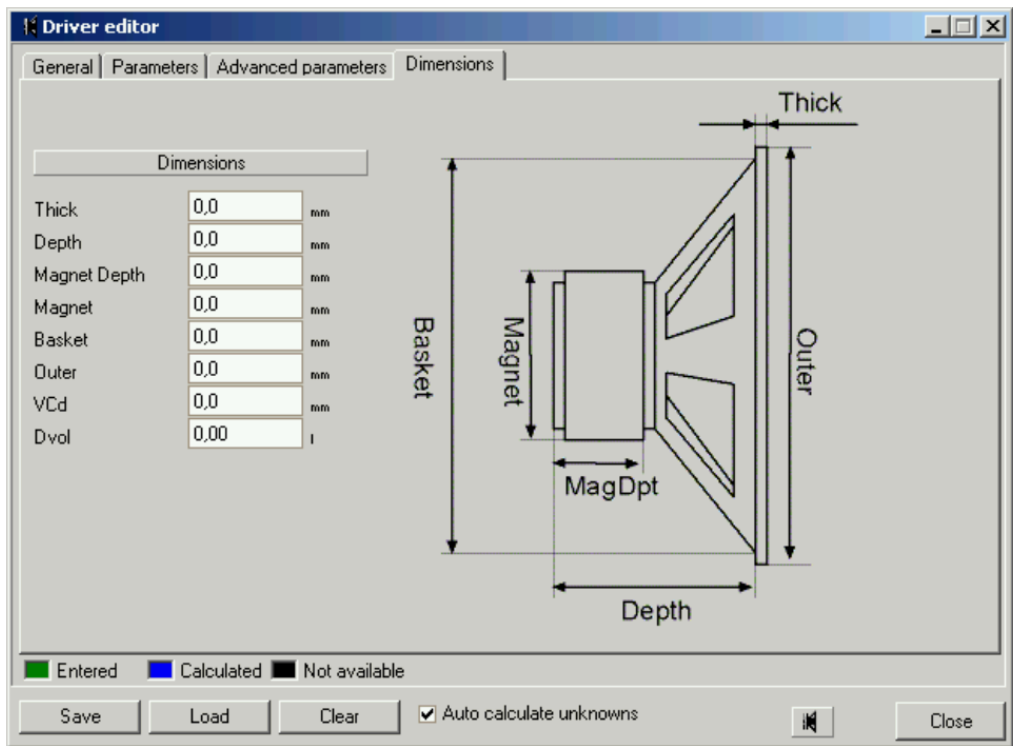
All of the drivers that you create are automatically saved in the \drivers subdirectory (folder) of your \WinISD directory (folder) and, as such, will not be deleted or overwritten when you update the WinISD database from the WinISD home page. The driver name that you provide will be the actual name of the file in the \drivers directory, so don't include any \s or anything else in the driver name that would be illegal as a Windows file name.

The bare minimum information that must be entered is the last three fields of first subsection (Qts, Vas, and Fs). Note that you won't get anything else but minimum set of graphs with these three parameters, so be sure to enter all relevant parameters. As you enter the parameters, driver editor will continuously check if new parameters can be calculated from parameters, which you have so far entered.

Parameter status is indicated by different colors. Green means user entered parameter, Blue means calculated parameter. Black means not entered nor calculated parameter. You can change entered parameters anytime, and the editor will re-calculate all calculated parameters all over again.

Important note for 0.50a7 and above:

The driver editor won't accept erraneous entries anymore. Your data must be perfect (i.e. not over-defined by entering "too much" parameters), for you to be able to save it. If you try to enter too many of them, editor will display an error dialog when you try to save the data.



In above tab, you can enter dimensional information of your driver. It shows then approximate displacement volume DVol. It is probably most accurate that can be obtained using just these dimensions. Most evident neglected volume is the driver frame. On the right, there is illustration for what most dimensions mean. Only one which is not illustrated, is the voice coil diameter, VCd. See also [Thiele/Small-parameters](#) section in articles.

The suggested procedure for entering driver parameters is following (check first that "Auto calculate unknowns" option is checked):

1. Enter **Mms** and **Cms**

This gives **fs**. If either is not available, then enter **fs** and other parameter.

2. Enter **Sd**, **Bl** and **Re**

Now, you should get all but **Qms** (and **Qts**), **Vas**. Please note that Vas may not match exactly what is specified by manufacturer, because exact value of Vas depends on environmental parameters. See [FAQ](#).

3. Enter **Rms** or **Qms**.

Either one will do, although I tend to prefer **Qms** over **Rms**, because it can usually be measured in driver measurement procedures.

4. Enter **Hc**, **Hg** and **Pe**.

If **Hc** or **Hg** or either is available, then enter **Xmax** and optionally either **Hc** or **Hg** if available.

5. Enter number of voicecoils.

This procedure is most accurate. Also note that it also calculates true SPL (1W/1m) value. So it might not match the marketing SPL value, which is generally somewhat vague. Not in all cases, though.

6. Correct **Znom**, if necessary.

If there are several voicecoils, then you must be careful when entering parameters in that case, because many manufacturers give **Bl** in voice coils in series, because it yields double value for Bl against parallel connection. If driver manufacturer gives **Qes**, **Bl** and **Cms** or **Mms**, then you can check how **Bl** is specified. For that, you can enter following parameters to calculate **Re**: **Qes**, **Fs**, **Mms** or **Cms** and **Bl**. Connection mode can be changed by changing the combobox selection. The driver editor then converts

Bl and Re values accordingly.

Equivalently, you can check for **Bl** by entering:

Qes, **Fs**, **Mms** or **Cms** and **Re** (for desired connection mode).

If you enter resistance for parallel connection and get about half of advertised **Bl**, then you know, that **Bl** was specified that way.

Like in [Driver-tab](#) in project window, you'll probably noticed the driver icon in bottom of the driver editor window. you can drag the driver object into some project by just dragging the icon.

Worked example of entering driver data into program

For example, Let's enter a entry for Scan-Speak 21W/8555-00 driver, which has following specs:

Characteristic sensitivity (2.83V / 1m)	87 dB
Recommended frequency range	20 - 600 Hz
Free air resonance frequency, fs	20 Hz
DC resistance, Re	5.5 ohm
Voice coil inductance, Le	0.4 mH
Effective piston area, Sd	220 cm²
Moving mass incl. air, md	32 g
Mechanical loss, Rm	0.9 Ns/m
Force factor, BL	8.2 Tm
Equivalent volume, Vas	136 ltr.
Mechanical Q-factor, Qms	4.5
Electrical Q-factor, Qes	0.33
Total Q-factor, Qts	0.31
Excursion, lin./max.	±6.5 / ±12 mm
Voice coil diameter	42 mm
Rated power handling	100 W
Net weight	2.4 kg

So what specs should be entered?

Based on guidelines given above, let's enter fs, Mms, Qms, Re, Bl, Sd, Le, Xmax and Pe. If you have incorrect unit displayed, just click on it until it matches with driver data units. As you enter the values, note how driver editor continuously calculates missing fields.

After entering those specs as suggested above, we obtain following entries in editor window:

Driver editor

GeneralParametersAdvanced parametersDimensions

Thiele/Small parameters

Qes0,329Qms4,500Qts0,307Fs20,00Hz

Vas135,87l

Electro-Mechanical parameters

Mms32,0gCms1978,9um/HfRms0,89361N/mRe5,500ohm

BL8,20000TmDd167,4mmLe0,00mHfSd220,0cm²

fLe0,00HzKLe0,000000H²=grt(Hz)

Large-Signal parameters

Xmax6,5mm peakHc0,0mmHg0,0mmVd143cm³

Xlim0,0mmPe100,0W

Miscellaneous parameters

no0,3214%Znom8,000ohmUSPL88,86dB

SPL87,22dB

Voicecoils1

ConnectionParallel

EnteredCalculatedNot available

SaveLoadClear

☒ Auto calculate unknowns

Close

As you can see, we got ourselves some of the specs what weren't even specified by the manufacturer. Calculated SPL 1w/1m spec matches also quite well, which is not the case with some of the brands. If you compare calculated Vas against manufacturer's spec, you'll notice that it matches quite well. So we can be quite confident that our data is reasonably correct.

When you have entered all the specs, you can check the "Advanced parameters"-tab, and see some statistics of your entered driver (calculated from the parameters we just entered):

newdriver.html[29/05/2024 11:07:43]

Driver editor

GeneralParametersAdvanced parametersDimensions

Thermal parameters

AlfaVC0,00001000/KR(t)0,00000X/WC(t)0,00000J/K

Figure of merits

SPLmaxLF83,67dB

SPLmax104,22dB

Rme12,22545N/m

gamma256,25000N/(A*kg)

Mpow3,49649N/sqrt(W)

Mcost0,00000N/m

EBP60,80Hz

Gloss9,5541%

Environment parameters

c343,68m/s

rho1,20095kg/m^3

EnteredCalculatedNot available

SaveLoadClear

☒ Auto calculate unknowns

Close

There, we can see, that this driver is definitely not suitable for horizontal mounting, Gloss is almost 10 percent! Also, using single woofer like this doesn't certainly be an SPL subwoofer, maximum predicted closed box SPL at 20 Hz at maximum excursion is only 83.7 dB. You can also see the air properties, which WinISD used when it calculated missing fields.

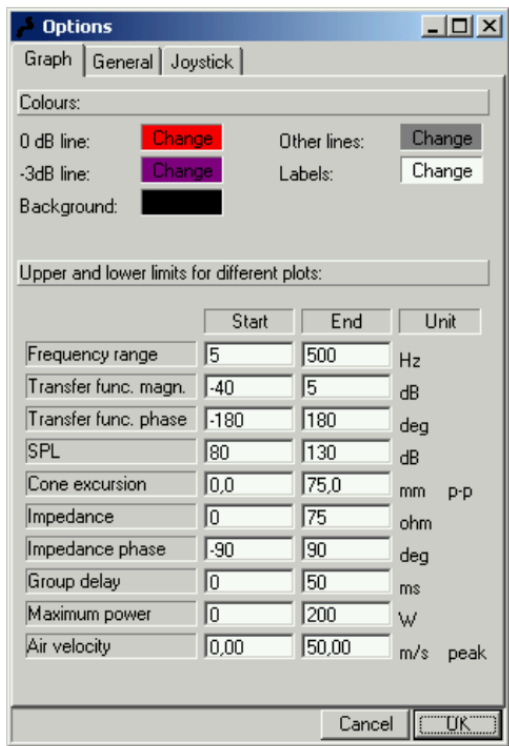
Finally, you can click "Save"-button to save the entered data. Driver editor suggests a filename for our driver.

This page originally written by JJ Richard (JJRichard@linearteam.org) and revised by Janne Ahonen (janne@linearteam.org)

Program Options

WinISD pro is quite configurable program. You can access this dialog by typing 'Alt+O' or with right mouse button while mouse cursor is over plot window. You can escape options dialog anytime by hitting 'ESC' (that is same than selecting cancel). Pressing enter means same thing than clicking on 'OK'.

Graph Tab



Here, you can change colors for various things, like plot background and grid lines. You can also set limits for various graphs. You can also set freely frequency range of interest, but please note that models used are quite useless, when it comes to high frequency issues.

General Tab

Options

Graph

General

Joystick

Username:

Janne Ahonen

Environment:

Temperature

20,00 °C

Relative humidity:

30,0000 %

Air pressure:

101325,0 Pa

Sound velocity

343,68 m/s

Desktop Grid:

☒ Snap to grid

X-Grid:

20

Y-Grid:

20

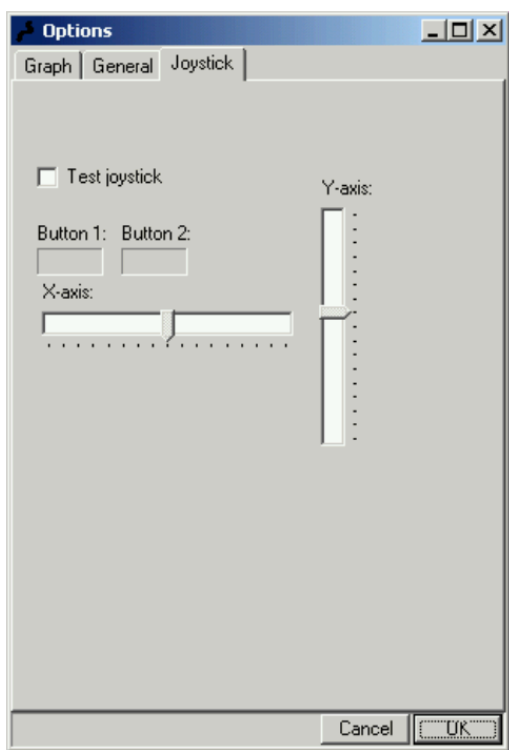
Cancel

OK

Environment default values can be changed here. Calculated sound velocity is also shown. These values are used every time new project is created, but you can change them per project-basis anytime you want. This is also unique feature of WinISD pro.

Here you can enter your name (which is used by default when entering new drivers to database). If you want, you can enable option to have desktop grid, which "snaps" the windows into grid.

Joystick Tab



Here you can test your joystick for signal generator usage.

Frequently asked questions

Since the release of WinISD, many emails have been received from folks pertaining to many issues of the program. While most of these issues are in fact discussed and explained in the Help File itself, most people (myself included) don't actually get around to reading Help Files unless we get stuck. So, with that in mind, I will continue to update the Help File with this FAQ section based on actual popular questions.

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1. General Questions

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 - [Why WinISD calculates different result for plot x with driver y in enclosure z?](#)
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 - [Why there is no option for nonlinear simulation?](#)
-

Q. [You had the EBP text bass-ackwards in the Help File.](#)
A. from JJ Richard (JJRichard@linearteam.org)

Yes, I most certainly did. Thanks to Choong for pointing that out to me. (5/26/99)

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Q. [How do I change the units of measurement from metric to standard etc....](#)
A: from JJ Richard (JJRichard@linearteam.org)

After you load a driver, simply click on the name of the unit over and over until your unit of choice is displayed. For example, if the unit is shown as 15 mm, then click on the "mm", and it will toggle through all available units! Easy, huh? WinISD will "remember" which unit you choose, and will subsequently default to that unit in the future. (5/26/99)

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Q. [What happened to vent mach calculator?](#)
A. From Janne Ahonen (janne@linearteam.org)

Vent mach calculator was replaced by much more useful tool, [vent air velocity graph](#), which gives not only the value of velocity, but also the frequency, where you could expect problems, if any.

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Q. Does WinISD add brace/driver/port displacements to calculated box volume?
A. From Janne Ahonen (janne@linearteam.org)

No. WinISD doesn't do guessing, and therefore shows **net volume** for your box. It would be hard to predict, how you are going to mount your driver. Or the port. Or how much and which kind of bracing you are going to use inside your box. For example, you could mount port completely external to the box, which case "usual" guess would be completely wrong. Bottom line is, that you must calculate and add any displacements by yourself to calculated box volume.

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Q. What calculated port length means? How it is measured?
A. From Janne Ahonen (janne@linearteam.org)

WinISD calculates port length assuming that there are one flanged and one free end, i.e. port is mounted to the wall of the box, and another end is freely inside the box. So end correction factor is 0.732 by default. Port length in box wall is counted to port length. If you want do convert the result to different end correction, then add 0.732 times diameter of the port to calculated port length. This gives you needed acoustical length of port. Then add end correction as you like. See also article about [port terminology](#).

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Q. Why driver editor calculates different value for Vas/Bl or some other parameter than in driver's manufacturer specs have?
A: from Janne Ahonen (janne@linearteam.org)

Most common reason why this is happening are the air properties. Since Vas depends on current air properties, there is no absolutely "correct" Vas value existing. So every time you'll adjust your room temperature, air properties will change, and your Vas will be different. My advice is to avoid entering Vas whenever possible. I also have personally found out that manufacturer's specs are not always in complete accordance with T/S model. So when in doubt, it is best that you verify parameters from driver manufacturer.

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Q. Why WinISD calculates different result for plot x with driver y in enclosure z?
A. from Janne Ahonen (janne@linearteam.org)

Most differences come from different input parameters. For example, vented box has quite a few other parameters than just box volume and tuning frequency. More complicated explanation is probably that calculation may use different model or different source parameters. If you go through all the relevant parameters, and still can't find the cause, please e-mail me. I'll have look at it then. After all, it is not at all impossible that we have made some mistakes somewhere.

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Q. Why frequency range is limited to low frequencies by default?
A. From Janne Ahonen (janne@linearteam.org)

That is limitation of used equivalent circuit models. They are valid only in piston range of driver, or about 200-300 Hz. Anything beyond that, is very inaccurate. Possibility to change the plot frequency range is provided just for convenience.

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Q. Why there is no option for nonlinear simulation?
A. From Janne Ahonen (janne@linearteam.org)

This is **not** a trivial topic as it may first seem. It is very hard to make accurate non-linear simulations, because for example considering the nonlinearities in the driver would require much more extra parameters to input to be able to simulate accurately. There are so many kinds of nonlinearities in drivers, that you would be running on the walls before you finished entering all the parameters. Hysteresis and thixotropic effects in spider/surround, inductance modulation due pole, Bl and Cms nonlinearity, you name it. Driver is not only source of nonlinearity, box air compression is adiabatic process. Vent air flow becomes nonlinear at

some point. And so on. Because there are no consensus, how the nonlinearities would be best described driver manufacturers won't generally give you any hint what kind and how much nonlinearity is expected from particular drive unit. Therefore, without the info, it would be just guessing, which is not good thing. There are actually some programs which calculate "nonlinear simulations", but if you don't have option to enter extra information about nonlinearities, the results are somewhat questionable. Yes, they are **closer** to the reality, but still they are inaccurate because lack of nonlinearity information. Furthermore, if the cone moves non-sinusoidally, then we'll have harmonics. Now, what should be plotted in SPL window? Just the fundamental or combined SPL considering fundamental and harmonics? Many kinds of questions arise.

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This page originally written by JJ Richard (JJRichard@linearteam.org) and revised by Janne Ahonen (janne@linearteam.org)

Box Types: The Great Debate

The debate over which design (Sealed vs. Vented vs. Passive radiator vs. Bandpass) is best will, like abortion, likely never end. Suffice it to say that no ONE design is best. They each have their Pro's and Con's. Anyone who tries to convince you otherwise is, at best, a well intentioned fool.

The keyword when considering which type of enclosure to use is...Trade-Off. There are trade-offs between box types, as well as trade-offs within box types.

Let's first begin with the Sealed box. This type of box is completely air tight, and uses the air trapped inside of the box as a suspension (like squeezing an empty plastic Coke bottle with the cap on) for the woofer. The "stiffness" of the suspension is determined by the size of the box in relation to the design of the woofer. A larger suspension (bigger box) will allow the woofer to move more freely, thereby yielding a lower bass response, but at a price in power handling, of course. Likewise, a smaller box will raise the low bass cutoff frequency, but will be able to withstand more power. There is a practical limit as to the upper and lower extremes here as well. You can achieve extremely low cutoff frequencies with sealed boxes, if you use so called Linkwitz-Transform equalizer.

Increasing the box size decreases the Qtc of the box. Lowering the box size raises the Qtc of the box. The suggested range of the Qtc is between 0.5 and 2.0 (the Qtc rating is listed in the Project Window under the Box tab). Overall, here's the scoop on the Sealed box:

- Pluses
 - Easiest of all boxes to design and construct.
 - Can better handle ultra-low (<30 Hz) frequencies without destroying the woofer.
 - Generally the most accurate box in the mid-bass frequencies (better transient response...notes in quick succession are more well defined).
 - Can be used with motion feedback.
- Minuses
 - Low bass response not as good as in a vented box.
 - Lower efficiency than that of a vented box.

Now, let's move on to the vented, or bass-reflex design. This design incorporates one or more tubes or "ports" which allow air to travel in and out of the box as the speaker cone moves in and out. The number of and size of the ports determines the -3dB point of the box. Frequencies which fall below the -3dB point aren't reproduced very well by the box. Here's the scoop on the vented box:

- Pluses
 - Better low frequency response.
 - Better efficiency.
 - Better power handling in vicinity of tuning frequency.
- Minuses
 - More difficult to design/build. If tuning frequency needs to be small, in small box, port length often becomes too awkward to realize properly. In such cases Passive radiator system may be favorable.
 - Frequencies below tuning can cause the woofer to overextend itself and therefore be damaged (an electronic low frequency filter can remedy this problem, so called 6th order alignment).

Passive radiator box is a variation of ported box. There, the mass is mounted in very compliant surround. Advantages and disadvantages over ported box are as follows:

- Pluses
 - Very low tuning frequencies can be realized in small boxes.
 - No port noise
- Minuses
 - Passive radiator may run out of excursion at low frequencies
 - As passive radiator has finite compliance, it creates notch in frequency response at passive radiator's resonance frequency. That notch may impair transient response.
 - Passive radiator unit is not normally suitable for horizontal mounting.

And finally, the bandpass box. This design must always be accompanied by a second set of speakers which will play the "mid-bass" frequencies. This is clearly the most difficult box to design and build, and is not intended for beginners. The bandpass box is designed to pass only a small range of bass frequencies. This design allows the user to select very specific woofers to meet very

specific needs. Each box/woofer design covers its own part of the spectrum which allows for a more even sound output across the entire spectrum as a whole.

The only real minuses to the bandpass design is that it is a monumental undertaking to properly design and setup a complete system successfully. I recommend you experiment with the sealed and/or vented designs first, and then go out and buy a book which deals with bandpass design/construction prior to building your first bandpass system. With your book in hand, use WinISD to do all the "what-if" scenarios for you. WinISD will go as deep as you want it to!

This page originally written by JJ Richard (JJRichard@linearteam.org) and revised by Janne Ahonen (janne@linearteam.org)

Crossovers: Passive and Active

Passive Crossovers:

Passive crossovers are installed in series with the speakers in your box, and serve to "funnel" the output frequencies from your amp to the appropriate speaker in your box. Sending low frequency notes to your tweeter will smoke it in no time flat. Sending high frequency notes to your woofer wastes high frequency power that would be better utilized by your tweeter. Your midrange fits in the middle there, and is subject to both scenarios.

Passive crossovers are constructed from capacitors and inductors:

Capacitors

Capacitors (caps) block low frequencies and pass high frequencies. Here's the electronic symbol for a capacitor:

-----| |-----

This is an accurate symbol in that it does a pretty good job of representing how a capacitor is actually built. Instead of getting into plates and dielectrics and all the rest, we'll just discuss how the thing functions. It's important to note that no electrical current flow actually passes THROUGH a capacitor. Think of a capacitor like this...imagine a horizontal pipe a foot long and about as big around as a coke can. Now, imagine that in the middle of the pipe is a rubber diaphragm (if you were to look into the pipe, you couldn't see all the way through..rather you'd see the rubber diaphragm in the middle of the pipe). Now, you couldn't pass anything (such as a smaller diameter pole, for instance) THROUGH this pipe from one end to the other. BUT, you could give the illusion of passing a pole BACK and FORTH through the pipe if your were to alternately push and pull on two different poles that actually met in the center of the pipe. As you push one way, the diaphragm will stretch a little, and then as you push the other way, it would stretch back to the other side. To an audience that didn't know any better, they'd swear that the pipe is indeed hollow!

This is how a capacitor affects electrical current. A direct current (DC) is completely blocked, while an alternating current (AC) is able to pass. Since DC is essentially ZERO Hertz, then it follows that maximum impedance is presented to LOW frequencies. The impedance drops lower and lower as the frequency increases (the faster you push/pull on the poles, the easier it becomes). Remember, a capacitor offers a "free ride" to HIGH frequencies.

Inductors

Inductors are the exact opposites of capacitors. An Inductor is a winding of wire which serves to block high frequencies, while giving the low frequencies their "free ride"! The electrical symbol for an inductor is:

__UUUUUU__ <---pretend like that is a non-broken line.

Any time current flows in a wire, an electric field is produced around the wire. When the current is then removed from the wire, the field "collapses". If you take a wire and wind it into a coil, things get interesting. As you initially begin to pass current through the coil, a field tries to build up around each winding in the coil (after all, it is in reality still just one length of wire). Trouble is, that since the coils are all so close to each other, they actually begin to interfere with each others progress! BUT, once a steady current flow is established in the wire, the fields around the coils quit fighting. As long as that current is kept constant, everybody is happy. When the current begins to change again, though (such as when the current is increased or removed), they get to fighting again, and continue to fight until the current quits changing. For this reason, inductors OPPOSE any CHANGE in current.

As far as DC and low frequencies are concerned, the inductor isn't much of a big deal. But for high frequencies which are always changing (and rapidly at that), the inductor is a bit of a pain in the ass. Remember, the inductor offers a "free ride" to low frequencies.

Caps and Inductors in action:

Ok, we know that the impedance of caps and inductors change with corresponding changes in input frequencies. The impedance that caps offer is called Capacitive Reactance (X_c), and is measured in ohms, given by the formula:

$$X_c = 1/(2\pi * F * C)$$

Where,

X_c = Capacitive Reactance (ohms)
 $2\pi = 6.28$
 F = Input Frequency
 C = Capacitance value of the capacitor itself (in Farads)

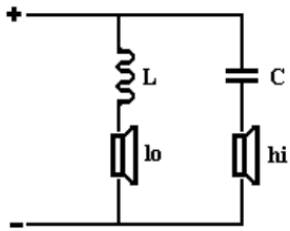
The impedance that inductors offer is called (surprise) Inductive Reactance (X_L), and is also measured in ohms, given by the formula:

$$X_L = 2\pi * F * L$$

Where,

X_L = Inductive Reactance (ohms)
 $2\pi = 6.28$
 F = Input Frequency
 L = Inductance value of the inductor itself (in Henries)

To put this in perspective, take a look at the following diagram of a typical 2 way passive crossover design (2- way means a woofer and tweeter only, no midrange driver):



Here, you see that the inductor is in SERIES with the woofer (marked "lo"), and the cap is in SERIES with the tweeter (marked "hi"). In keeping with what we discussed above, notice that the inductor (who gives the free ride to low freqs) is indeed connected to the woofer. This allows all the low, bass notes to travel to the woofer, and also blocks the high freqs from reaching the woofer. Likewise, the cap is connected to the tweeter, which gives the free ride to the high frequencies, while blocking those deadly low frequencies from reaching the tweeter.

Naturally, the next logical question is "Yeah, but what size cap and inductor do I use?". Good question. The first thing to determine, of course is what frequencies do you want to chop off?

If you want to stop all frequencies above, say, 1500 Hz from reaching the woofer, then you need to use an inductor that will offer the same impedance that the woofer is rated at, at 1500 Hz. For example, if you're using an 8 ohm woofer, and your cut-off frequency is to be 1500 Hz., then you need to choose an inductor whose impedance is 8 ohms at 1500 Hz!

Recall the formula for Inductive Reactance:

$$X_L = 2\pi * F * L$$

Well, we know that we want the inductive reactance to be 8 ohms at an input frequency of 1500 Hz. So, plugging that in, we get:

$$8 = 2\pi * 1500 * L$$

We need to get that L alone, so we divide both sides by $(2\pi * 1500)$:

$$8 / (2\pi * 1500) = (2\pi * 1500 * L) / (2\pi * 1500)$$

The $(2\pi * 1500)$ on the right side of the equation cancels itself out:

$$8 / (2\pi * 1500) = L$$

Ah, that's better. Breaking out the trusty calculator (or use the built in calculator in Windows by clicking START, RUN, and type in CALC, and hit ENTER), we get:

$$L = 0.0008493 \text{ Henries.}$$

Move the decimal 3 places to the right to get the answer in the more popular milli-Henries (mH):

$$L = 0.849 \text{ mH}$$

Now, suppose we want the same cutoff frequency for our tweeter. The same rule applies..we need a cap that will offer 8 ohms (provided the tweeter is an 8 ohm tweeter, of course) at 1500 Hz. Looking at the formula for capacitive reactance (X_c) again:

$$X_c = 1/(2\pi * F * C)$$

We get:

$$8 = 1/(2\pi * 1500 * C)$$

We need to get that C alone, so begin by taking the inverse of each side:

$$1/8 = (2\pi * 1500 * C)/1 \text{ <---the 1 cancels itself since any number divided by 1 is itself.}$$

Which gives us:

$$1/8 = (2\pi * 1500 * C)$$

Dividing both sides by $(2\pi * 1500)$ frees up the C:

$$(1/8)/(2\pi * 1500) = C$$

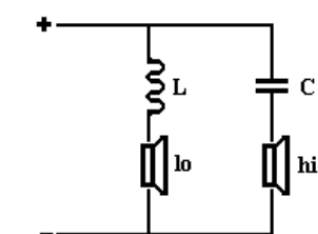
Ah. Now for the calculator:

$$C = 0.00001327 \text{ Farads}$$

Moving the decimal 6 places to the right gives us the answer in the popular micro-Farads (μF):

$$C = 13.27 \mu\text{F}$$

Ok, so our cap is gonna need to be 13.3 μF , and our inductor needs to be 0.849mH in the crossover. Look again at the diagram:



At 1500 Hz input frequency (from the + and - terminals in the diagram), our inductor, L, will offer 8 ohms of reactance (resistance). Since the woofer is also rated at 8 ohms, this means that the voltage being applied to the woofer is half of what it would be if the inductor weren't there. This results in a 6 dB loss from the woofer at 1500 Hz. And remember, inductors hate high frequencies! So that means that when a higher frequency comes along, the reactance offered by the inductor will be even HIGHER than 8 ohms. In fact, for every octave (doubling of) increase in frequency, the inductor will lower, or attenuate, the output from the woofer by an additional 6 dB! This is referred to as a "6 dB/Octave" crossover (plug in other frequency values into the inductive reactance formula and you can see how much reactance the inductor offers to the circuit at those frequencies...remember to use 0.000849 as the value for H)....

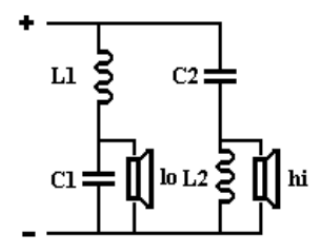
Likewise, at 1500 Hz, the cap also offers 8 ohms of reactance, which cuts the output from the tweeter by 6dB as well. And, since caps block low frequencies, each octave (doubling) of frequency drop will cause a corresponding 6 dB of attenuation to the cap as well. Again, plug in different values to the capacitive reactance formula and look at the different impedances offered into the circuit by the cap at different freqs (remember to use 0.0000133 as the value for F).

You can use these formulas to figure out which size caps or inductors are needed to cutoff any speaker at any frequency.

Going further:

Sometimes, cutting off a speaker at a rate of 6 dB/Octave isn't enough. At 6 dB/Octave, especially in high power systems, alot of lows are still powerful enough to get through and damage the tweeters. In this case, a 12 dB/Octave crossover can be used to

"steepen the curve" of attenuation. In this case, the actual cutoff frequencies are still the same, but for each corresponding octave change in frequency, the input to the respective driver is reduced by 12 dB, instead of only 6 dB. How? Look at the following diagram:



In this design, BOTH the woofer and the tweeter are connected to an inductor AND a cap! We'll stay away from the long math (just let WinISD [figure it out](#) for you). However, a quick explanation will clear it up for you.

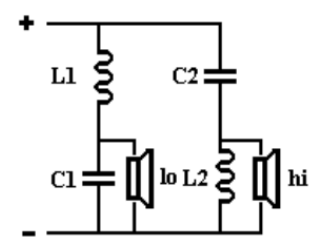
First, let's look at the woofer. Inductor L1 is in SERIES with the woofer. This means that any and all frequencies that pass through the woofer MUST also pass through L1. Well, L1 works just like he did before. He blocks the highs from reaching the woofer. But, since L1 can only block 6 dB/Octave's worth of high freqs, there are still some high freqs that sneak through. Here's where cap C1 comes into play. Notice that C1 is wired in PARALLEL to the woofer. This means that the cap can shunt, or bypass, frequencies around the woofer. Remember, capacitors give high freqs a "free ride". So, any high freqs that sneak through L1 are given a shortcut by C1. Why should the high freqs bother going through the woofer when C1 is giving them a "free ride" straight over to the other side? What it all boils down to is that L1 does his 6 dB/Octave's worth of work, and C1 does an additional 6 dB/Octave's worth of work. Together, they combine to yield a net of 12 dB/Octave of attenuation!

Capacitor C2 and Inductor L2 work exactly the same routine on the tweeter. C2 blocks as many lows as he can, and any lows that sneak through are given a free ride by L2.

Notes and Caveats:

Inductors and capacitors only come in certain values. Our circuit called for a 13.3 uF cap. You're not going to find any 13.3 uF capacitors. However, 10 uF caps are quite common, and using a slightly different value won't kill you. Also, the same is true for inductors. In fact, you don't have to put a crossover network on your woofers...

12 dB/Octave passive crossovers can be dangerous to your amp. Look again at the 12 dB/Octave crossover:



Notice that in each branch, the inductors and capacitors are in SERIES with each other (L1 is in series with C1, and C2 is in series with L2). In each branch, the inductor and capacitor are each chosen so as to offer 8 ohms of reactance at 1500 Hz. What this means is that at 1500 Hz, the cap is offering 8 ohms of capacitive reactance, and the inductor is offering 8 ohms of inductive reactance. Capacitive reactance and Inductive reactance are exact opposites, and cancel each other out! If you have 23 ohms of capacitive reactance, and 18 ohms of inductive reactance in a circuit, the net reactance is 5 ohms (the difference of the two). If the woofer or tweeter are removed from the circuit (become unplugged, or electrically "open" during play), then what you have is 8 ohms of capacitive reactance, and 8 ohms of inductive reactance in that branch. Since the two reactances cancel each other out, then the amplifier effectively sees a grand total of ZERO ohms at 1500 Hz! This will result in a damaged amp. Keep this in mind when considering implementing a 12 dB/Octave passive crossover into your system. Amps with built in short circuit protection will most likely survive this scenario, however...

Active Crossovers:

Active crossovers are used mainly on multi-amp systems. Active crossovers are small, electronic circuits which serve to separate frequencies at the "line-level", and send those frequencies to separate amps. Each amp then plays the frequencies it's given. Each amp then powers it's own special set of speakers (one amp for woofers, another amp for tweeters, etc.)

[WinISD](#) will figure out the values you need to construct active crossovers, however, the in's and out's of the design of these devices is way beyond the scope of this Help file. If you can't look at the diagrams and immediately recognize what components you need to buy and solder together, then don't even attempt it. Go buy one from the store!

A good source for active filter design is:

Active Filter Cookbook (2nd Edition)
Don Lancaster
ISBN 0 7506 2986 X (Newnes)
(c) 1996

This page written by JJ Richard (JJRichard@linearteam.org) and revised by Janne Ahonen (janne@linearteam.org).

dB's, Octaves, and Frequency (Hertz)

What the hell is a decibel, and what the hell is an octave?

An octave is any doubling or halving of a given frequency. A dB (decibel) is a unit of measurement which is "referenced" to some value. I believe that the official definition of a decibel is "A logarithmic means of measuring a gain or a loss". Decibels are funny, and a little hard to understand at first. Decibels apply to sound loudness, power, or voltage. The best way to think of decibels is to think "compared to what?". When speaking of sound (loudness), a 1 dB change is the minimum amount of change perceptible by the human ear. If you were sitting blind folded in my living room, and I snuck over the volume knob and slowly changed it, the point at which you would say "Ah! I can tell you've changed the volume!" is a 1 dB change in sound. If I continue to "crank the knob", I would have to change the sound level by 10dB to make the sound twice as loud as it was from where it started. This is true from any volume level. Decibels simply define a ratio of change from "what was" to "what is" in either watts, volts, current, or SPL. A 3dB increase of sound is about the increase you make when you turn the volume up to the next reasonable level, like when someone says "turn it up a little, I can't hear it". The price of bringing the volume "up a notch" --- twice the power.

Watts:

Here's a little secret...if you see a 100W amp next to a 130W amp in a store, those 30 "extra" watts are meaningless as far as loudness is concerned. Those extra 30 watts may provide a small amount of reserve power (useful for reproducing deep bass notes that happen in rapid succession), but a properly designed amp will have large enough capacitors to fulfill that need anyway. Remember that 3dB I told you about? Well, it actually takes a DOUBLING of power to reach that 3dB! In other words, if you're cranking out 100W of power from your amp, and you go to the volume knob to "crank it up a notch", you will actually need to DOUBLE the power output to increase the volume up enough to take it to the next "level". That's right, to increase the volume so that the increase is plainly noticable requires a DOUBLING of amplifier power! So, your 100W amp is probably every bit as good as that 130W amp! If you have a 100W amp, and want a LOUDER amp, you must at LEAST double your power. You need a 200W amp! If you want the sound to be "twice as loud", then you must increase your power tenfold! A 1000W amp will only be twice as loud as a 100W amp! So the next time a salesman tells you that those extra 30 watts will make the difference, you can tell him to pack sand (in fact, go ahead and ask him about "reserve power"). ;)

Frequency:

The human ear can detect (in young people) a frequency range of 20 Hz to 20,000 Hz. In reality, the average person can hear frequencies from about 28 Hz to 18,000 Hz. And to top that off, the average musical song will only cover frequencies from 35Hz to 15,000 Hz.

For you "need to know" types, Frequency correlates to the actual movement of a given speaker. Imagine placing a D cell battery across the terminals of a speaker. The speaker would respond by moving either in or out with a little "pop". Reversing the polarity of the D cell battery would cause the speaker to now move in (if it moved out the first time) or out (if it moved in the first time). If you were the 6 Million Dollar Man, and could switch the battery back and forth at incredible speeds, the speaker would start to make sound. If you could switch the battery back and forth at a rate of 30 times per second, the speaker would play a 30 Hz note (albeit with an insane amount of popping as the battery continued to make and break contact, but you get the idea)...and so on.

Passive Crossovers:

Passive crossovers serve to "funnel" the output frequencies from your amp to the appropriate speaker in your box. Sending low frequency notes to your tweeter will smoke it in no time flat. Sending high frequency notes to your woofer wastes high frequency power that would be better utilized by your tweeter. Your midrange fits in the middle there, and is subject to both problems.

Passive crossovers are installed in series with the speakers in your box, and offer varying amounts of impedance (A/C resistance) to the amplifier. A capacitor's impedance is very high at low frequencies, and very low at high frequencies. Placing a capacitor in series with your tweeter will serve to "block" the very low frequencies, while allowing the very high frequencies to pass almost for "free".

Hearing Loss (you're only young once):

An outward moving speaker increases the air pressure in the room, whilst an inward moving speaker decreases it. This rise and fall of air pressure in the room pushes little hairs inside of your ears back and forth (imagine tall weeds in a grassy field being pushed by the wind...only not just any wind, but a strange wind which blows one way for an instant, and then the opposite way the next instant, continuously). These little hairs in your ear perceive these changes in the "wind", and pass this onto your main

noodle, which processes this as sound.

These little hairs, however, can only move so fast. At frequencies above about 18 kHz (or below about 25 Hz), the little hairs can't dance...and this is the reason why you don't hear sounds outside of that frequency range. Also, if the INTENSITY of the "wind" is too great for too long (you "jam out" at mind blowing volume levels all the time), those little hairs begin to get "stuck" in the bent over position (just like the weeds in the field would naturally do if the wind got high enough). When this happens, you lose the ability to detect sound.

At first, the high frequencies vanish (the tallest, most sensitive weeds go first), and eventually all of them "die". There is no cure at this time for this hearing loss. Hearing aids are available, but they only serve to CRAM the now defunct freqs down your ears' "throat", and is the equivalent of walking on crutches. Bottom line is this...if your ears ring right after you've been jamming out, then you've just killed some weeds (in your ear ;)). I myself have fallen victim to this trap (thanks to too much AC/DC, Frampton, RUSH and the like).

So, a word to the wise (to those who are smart enough to heed it). Your average senior citizen on the street today has a hard time hearing, and he wasn't exactly jammin' out to a Rockford Fosgate Punch amp when he was young, right? So, here's something to dwell on..."Ya ain't gettin' any younger"....and neither are your ears...you WILL age, and you WILL live to regret any abuse you subject your ears to... Make it "thump", not BLARE! If your friend is showing off a system, and it's either distorting OR blaring loud, then promptly stamp a big, fat "L" on his forehead and move on. Keep your sound in the "comfort zone".

'Nuff said.

This page originally written by JJ Richard (JJRichard@linearteam.org) and revised by Janne Ahonen (janne@linearteam.org)

Port terminology: Physical/acoustical length, end correction

If you have read articles on port design on the Internet, you have probably stumbled on various terms involving ports. This document is to explain some of those terms.

Effective radius/diameter

If your port has some other shaped cross-section than circular, then we'll need to calculate effective radius or diameter. Effective radius is radius of the circle, which has same cross-section area than non-circular object. For ports, only inside area contributes to port area, not it's walls. For calculating effective port radius/diameter, use following formulas:

effective radius = $\sqrt{S/\pi}$ and
effective diameter = $\sqrt{4 \cdot S/\pi}$,
where sqrt means square root.

For example, you have square port which has 100 square inches of cross-section area, what is effective diameter and radius?

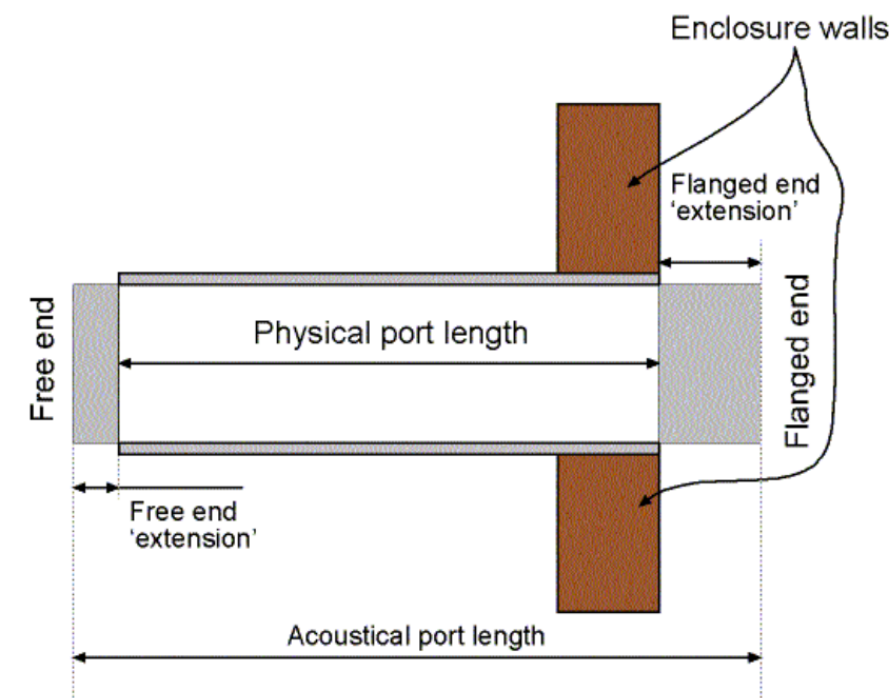
Now, just plugging the values to above formula we get:

radius = $\sqrt{100/\pi} = 5.642$ inches
diameter = $\sqrt{4 \cdot 100/\pi} = 11.284$ inches.

Of course, diameter could have been easily obtained from radius by just multiplying radius by two.

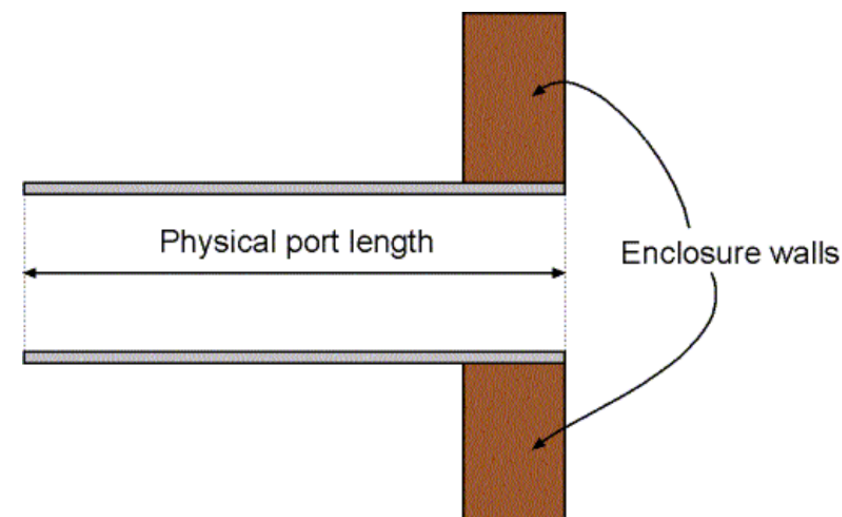
Acoustical length

First, we need to discuss what is the idea behind port, and how it works. Basically, port forms a mass-spring resonance circuit with enclosure volume compliance. Helmholtz was first to discover such a resonator, hence it was named after him. Mass in port resonates with compliance of box air. By tuning the enclosure, we chose mass in port, so that resonates at desired frequency. As our port will have some area, we can calculate, how long this "tube" should be to have desired mass. This length is so called "acoustical length" of our port.



Physical length

Unfortunately, effect of the port doesn't stop at the end of our port. Moving mass extends somewhat beyond of the port. Researchers of acoustics say that radiation impedance has reactive component, mass. So if we make our port length equal to calculated acoustical length, we'll find out that tuning frequency is lower than what was expected. We can make good approximation to this extra mass by adding this radiation load mass from desired port mass and then calculating required port length.



By default, WinISD assumes that you are going to mount your port above way. And therefore gives you physical length of your port assuming mounting like in above figures. See also [FAQ 3.1](#).

End correction

End correction factor is analytically or empirically determined factor, how much port extends beyond its physical ends. For free end, end correction is 0.30665 times port effective diameter. For flanged end, more analytical expression is available, $4/(3 \cdot \pi) \sim = 0.42441$ times port effective diameter. Flanged end is calculated assuming that tube terminates to infinite baffle. Which is not exactly true, but.. Following table summarizes various port configuration types and their total end correction factors (refer to above picture to various end types):

Summary of end correction values			
Port configuration	end correction 1	end correction 2	Total end correction
Two free ends	0.30665	0.30665	0.6133
One flanged and one free end	0.42441	0.30665	0.7311
Two flanged ends	0.42441	0.42441	0.8488

Physical port length is obtained by subtracting port effective diameter multiplied by suitable end correction value from port's acoustical length.

Thiele/Small Specs - What They Mean

The Thiele/Small specs that are constantly being referred to throughout these help files are the product of two Australians...namely A.N. Thiele and Richard Small, who, back in the late '60's and early '70's, applied filter synthesis methods to box design. Before that, it was process of trial and error and some rules of thumb!

These two gentlemen deserve some more publicity, so I'll include photographs of theirs into this help file. Here are the personal photographs of Nevielle Thiele and Richard Small:



Neville Thiele Richard Small

They accomplished this by equating a woofer in a box to an electronic low or high pass filter. While they weren't the first to discover the aforementioned analogy, they were indeed the most prolific at it. So much so, that even today, speaker designs are almost exclusively based on their research, math, and formulas...ie, the Thiele/Small Specifications that you hear so much about.

Here is explanation to various driver parameter encountered in driver editor (from upper left to bottom right):

Thiele/Small-Parameters section

- Qes** Electrical Q (=damping), lower value means higher damping. It describes a drivers ability to resonate at fs based on electrical means.
- Qms** Mechanical Q (=damping, lower value means higher damping). It describes a drivers ability to resonate at fs based on mechanical means.
- Qts** Total damping (parallel coupling of Qms and Qes).
- Vas** Equivalent Volume of air to Cms.
- Fs** Free air resonance frequency of driver.

Electro-Mechanical parameter section

- Mms** Mechanical Mass of the vibrating part of the driver including air load.
- Cms** Compliance of driver (inverse of spring stiffness).
- Rms** Mechanical damping, gives you the mechanical damping of the diaphragm arising from mechanical friction, including the resistive part of the radiation load. Rms can be compared to Rme, and Rms is similarly related to Qms. Larger Qms gives

	smaller Rms. For woofers this is normally desired because the suspension then operates closer to a perfect spring.
Re	DC-resistance of voice coil.
BL	Magnetic Induction crossed with wire length in the airgap. (crossing = cross product, a mathematical vector-operation)
Dd	Diameter of Diaphragm.
Le	Voice coil inductance.
Sd	Surface-area of Diaphragm.
fLe	The frequency at which Le and KLe is to be determined.
KLe	Voice coil semi-inductance in $[H \cdot \sqrt{Hz}]$, after Vanderkooy

Large-Signal Parameter section

Xmax	Max linear excursion, usually calculated as $\text{abs}(H_c - H_g)/2$, and sometimes multiplied by a factor (1.15 or 0.87, depending on how much distortion is accepted). WinISD does not multiply with any of these factors. Some manufacturers erroneously gives you Xmax as the damage limit, see Xlim. WinISD needs this as peak value.
Xlim	Damage limit excursion, also a peak value.
Hc	Height of coil.
Hg	Height of airgap.
Vd	Volume Displacement, how much air the driver can move in its linear range.
Pe	Thermal limited max. continuous electric power handling. If a driver is driven continuously above Pe, then it will eventually fail.

Miscellaneous parameters section

no	Efficiency (n should be the greek letter "eta") in percent [%].
Znom	Nominal impedance of the driver (not used in simulation).
USPL	efficiency in deciBell (SPL = sound pressure level) in $[dB/2.83V]$ dB per 2.83 Volt (similar to 1W into an 8 ohm load). This SPL-measurement is similar to SPL (see above), but gives different values. This shows you the difficulties about matching drivers. With 8 ohm drivers 2.83 Volt gives you 1 Watt and the two figures (SPL and USPL) will be similar, but at lower impedance levels the USPL level will increase. USPL is the socalled voltage sensitivity and is closer to application with voltage amplifiers. To a limited extent you could match drivers for a loudspeaker system with this factor.
SPL	Efficiency in deciBell (SPL = sound pressure level) in $[dB]$ per Watt directly related to no, but definately not an "accurate" figure in applications. In other words, if a speaker driver is specified by the manufacturer to some other value, do not use that value for WinISD unless you need it to calculate some Thiele-Small parameters and approximate values are better than no values at all. WinISD assumes distance 1 meter, radiation into halfspace ($2 \cdot \pi$), and voltage driv. SPL is the so-called power-sensitivity, not really related to application, normally voltage amplifiers are used, but can become relevant if you want to compare two similar drivers with different nominal impedance levels.
Voicecoils	Voicecoils is a descriptive parameter. It just tells how many voicecoils there are. So ordinary drivers have 1, DVC drivers have 2 etc.

Thermal Parameters Section

AlfaVC	AlfaVC is resistance temperature coefficient of voice coil material. It tells what is relative change of voice coil resistance per unit of temperature. Expressed in 1/K. Normally, copper has AlfaVC of about 0.0039 1/K at +20 °C.
R(t)	R(t) is thermal resistance of driver from voice coil to ambient box air in Kelvins/Watt (K/W). This is not used yet in simulations.
C(t)	

C(t) is thermal capacity of driver voice coil assembly in Joules/Kelvin. This is not used yet in simulations.

"figure of merits" parameters section

- SPLmaxLF**
SPLmaxLF gives how loud driver can play in closed box or infinite baffle into half-space at maximum excursion at 20 Hz. Distance from this imaginary baffle is 1 meter. It gives "feeling" on Vd. Note also, that it doesn't apply to vented or any other assisted enclosure.
- SPLmax**
Maximum thermal limited SPL in [dB] (at maximum Pe, assuming power compression = 3 dB) playing into 2pi space
- Rme**
Electromagnetic Damping Factor in [N·s/m] (the unit for viscosity), gives you the mechanical control/damping of the diaphragm arising from the electro-magnetic motor system. Rme is related to Qes in a way similar to how Rms is related to Qms. Rme is often used as a measure for power of the magnetic motor system, see Mpow and Mcost.
- Gamma**
The acceleration factor (acceleration per ampere) in [m/(s²·A)].
- Mpow**
Motor power-factor in Newton per square-root Watt [N/sqrt(W)]. Similar to Mcost. I have seen Rme as a measure of motor power, but this is simply the square-root of Rme, and it provides a simple measure in Newton, which I (**Claus Futttrup**) prefer, and which seems to relate the actual (subjectively perceived) power in a linear way. The square-root Watt unit can be difficult to understand, but should be interpreted as square-root of Volt * Ampere. In this respect it becomes clear that Mpow is independent of the drivers impedance level, and therefore does not prioritize high or low impedance drivers. Mpow is purely a motor system power-factor.
- Mcost**
Motor cost-factor in [N·s/m] (or [kg/s]). Mcost expresses how powerful the motor system is (based on Rme, Xmax and either Hc or Hg depending on whether the voice coil is overhung or underhung), and the Xmax value includes an indicator of how much efficiency is "lost" in the design. This factor is therefore a description of how expensive the motor system is. This is an indicator on the price of the driver, but please forget about the unit. Other factors comes in, like diaphragm material, manufacturing tolerances etc. This version of Mcost (instead of using Rme) is based on an extension suggested by T. L. Clarke, where the cost of getting a high Bl at low impedance must be even higher when the driver is significantly overhung or underhung.
- EBP**
Efficiency-Bandwidth-Product in [Hz]
- Gloss**
Gloss tells how many percent of Xmax driver cone will sag, if driver is mounted horizontally. This figure is only approximate, because surrounding air shifts resonance frequency to somewhat lower value. So result is somewhat over-pessimistic by some amount. Generally, figures of more than five percent tell you that driver in question shouldn't be mounted horizontally. Value of gravity acceleration used when calculating sag is 9.80665 m/s².

Dimension parameters section

- Thick**
The thickness of the basket plate.
- Depth**
Depth of the driver.
- Magnet Depth**
Magnet depth/height/thickness (cylinder height).
- Magnet**
Magnet diameter.
- Basket**
Basket diameter (the hole to cut in the baffle).
- Outer**
Outer diameter (the space to make room for on the baffle for the driver).
- VCd**
Voice coil diameter.
- Dvol**
Driver displacement volume. Approx. the box volume occupied by the driver, when mounted with magnet pointing into the box.

Ultimately, speaker design is more of an art than a science, and as such you could dig up, process, and ponder a lifetime of math, specs, and equations when designing a speaker. In fact, you could spend 1000 years designing, calculating, and crunching numbers in an attempt to create the "perfect" speaker. However, even after spending that 1000 years of planning, somebody would still find something "that could be better", about your box!

Therefore, I will not even make an attempt to cover "all" of the speaker design technology here (not only am I under-qualified for such a task, the very field itself lends itself to constant controversy which I refuse to lose sleep over).

WinISD was not written/developed as a "this is it, no matter what you say" program. Instead, it is intended to give even the casual audiophile a means to build a GREAT sounding box (using the available technology).

Just use the program as it is intended to be used, and enjoy the results, because you WILL end up with one Helluva great sounding box. Leave the constant bickering and "one upsmanship" alone!

*This page written by JJ Richard (JJRichard@linearteam.org) and revised by Janne Ahonen (janne@linearteam.org)
Explanation of different driver parameters originally written by **Claus Futtrup** and used here with permission.*

Introduction to equivalent circuits

These are the models which WinISD graph calculations are based on. Models are documented here for reference. Most box-program authors are not documenting publicly these things. They are also useful, if someone wishes to evaluate validity of used models. Most people don't need these.

Circuits are reduced to acoustical side. Models for currently supported enclosure types are:

- [Closed box](#)
- [Vented box](#)
- [Passive radiator](#)
- [4th order bandpass](#)
- [6th order bandpass type A](#)

Voltage source, U_{ad} , is so called "constant pressure generator". Resistance models acoustic resistance, capacitor models compliance (volumes) and inductor models mass. There are several ways to determine component values. One, which is used in WinISD pro, is where we start on driver V_{as} to determine C_{cas} :

$$C_{cas} = V_{as} / (\rho \cdot c^2)$$

Then, we'll use f_s to determine L_{mas} :

$$L_{mas} = 1 / ((2 \cdot \pi \cdot f_s)^2 \cdot C_{cas})$$

Then, Q_{es} and Q_{ms} is used to determine R_{ae} and R_{am} :

$$R_{ae} = 1 / (2 \cdot \pi \cdot f_s \cdot Q_{es} \cdot C_{cas})$$

$$R_{am} = 1 / (2 \cdot \pi \cdot f_s \cdot Q_{ms} \cdot C_{cas})$$

Now, we'll calculate box volume modelling capacitor C_{cab} just like C_{cas} . If we have closed box, we'll have only determine the leak and absorption modelling resistors. These are determined by so called Q-factors. For lack of better models, these are determined at resonance frequency of boxed driver.

Determining frequency response

When all component values are in place, then we'll just simulate completed circuit, using any circuit simulation program, e.g. SPICE (**S**imulation **P**rogram with **I**ntegrated **C**ircuit **E**mphasis). Frequency response is calculatable from determining the volume velocity through C_{cab} . Using one well-known relation, we can get to far-field pressure value from volume velocity:

$$p(r) = \rho \cdot s \cdot U_0(w) / (2 \cdot \pi \cdot r) \cdot \exp(-j \cdot k \cdot r)$$

where k is so called wave number ($k = w/c$), and r is distance from diaphragm. U_0 is total volume velocity produced by box. If there is only one chamber, then U_0 is current through C_{cab} . If there are several chambers, like in bandpass enclosures, then total volume velocity is **vector** sum of all volume velocities of chambers. Latter $\exp()$ -part is the phase factor (which has always gain of unity), which is neglected in WinISD, in order to show more clear graph. Pressure is therefore complex valued. Now, when we have obtained absolute pressure level, we may use reference pressure of 20 μPa , the 0 dB reference.

Group delay

Group delay can be extracted from frequency response data by numerically derivating the phase by mathematical definition: $gd(w) = -d(\arg(H(f))) / dw$, where $\arg(H(f))$ is phase part of transfer function. In practice, it is done by just calculating two phase values on very small interval, and then just approximating above differential by $(\arg(H(f+d)) - \arg(H(f-d))) / (2 \cdot d)$.

Cone excursion

Cone excursion is calculated by first determining volume velocity through driver. Then, volume velocity is integrated, and the result is then divided by S_d to get cone excursion. Still, we need to scale result by $\sqrt{2}$ if we want to get peak value and $2 \cdot \sqrt{2}$ if we want peak-to-peak value.

Port velocity

Port velocity is like cone excursion. Again, volume velocity through port is determined. Since we need port velocity, then, we just divide volume velocity by port area. This gives us RMS port velocity, which is optionally scaled, like cone excursion.

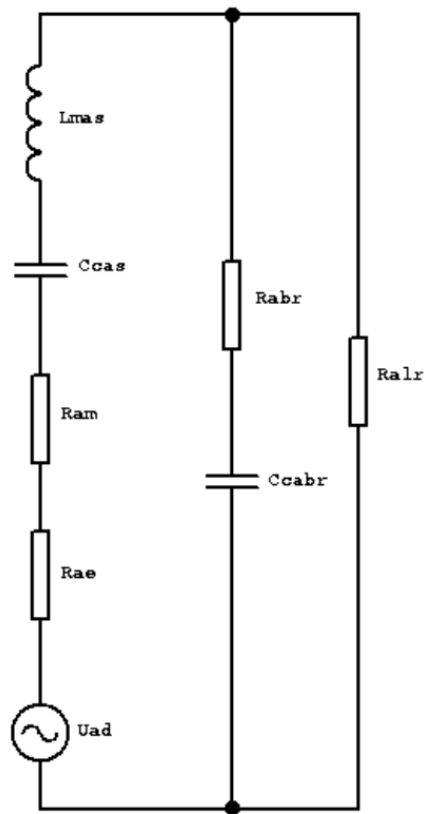
Impedance

Impedance is calculated by first calculating acoustical impedance seen by Uad and Rae is subtracted. By using conversion formula, $Z_{em} = Bl^2 / (Sd^2 \cdot Z_a)$, where Z_a is aforementioned acoustical impedance and Z_{em} is additional motional impedance reflected from mechanical circuit. Then final electrical impedance Z_e is calculated by $Z_e = R_e + j\omega \cdot L_e + Z_{em}$. All impedances are complex.

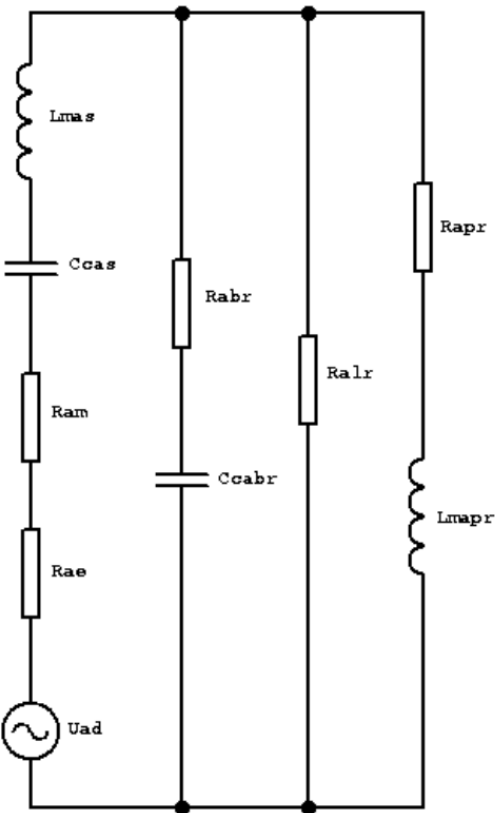
This concludes this brief introduction how simulations work in WinISD internally. For more detailed description, there are many good books about the subject. One good one is written by W. Marshall Leach, Jr.:

*Introduction to electroacoustics and audio amplifier design
W. Marshall Leach, Jr.
Kendall/Hunt publishing company
ISBN 0-7872-7861-0*

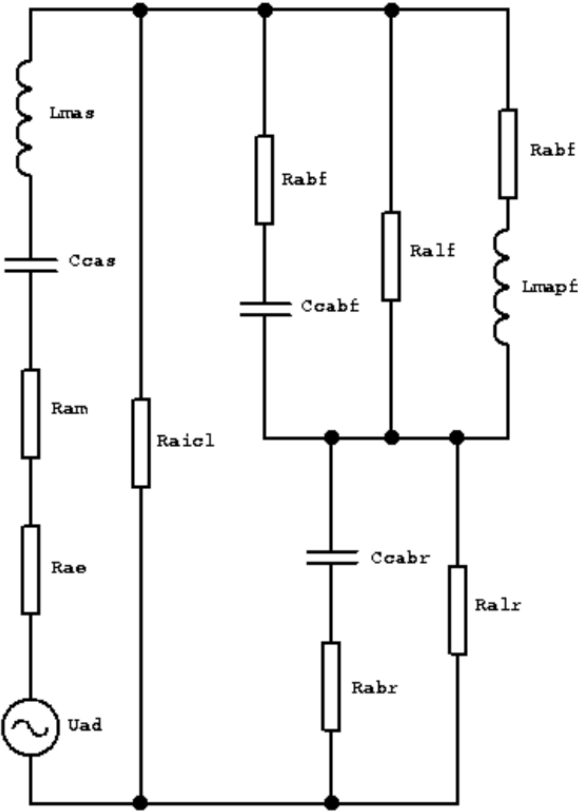
Closed



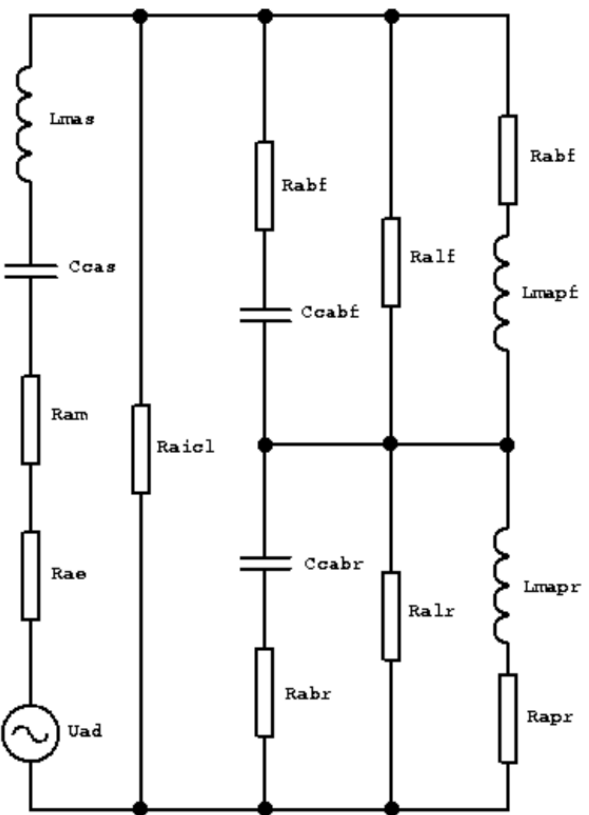
Vented



4th order bandpass



6th Order Bandpass type A



Passive Radiator

